ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

85 question sets • 2555 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-49)}$ in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. $\sqrt{(-49)} = \sqrt{49} \cdot \sqrt{(-1)} = 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write √(−18) in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

3√2·i

GRADING NOTE Form required: simplest radical form. √(−18) = √9 · √2 · i = 3√2 i. Not accepted: √18 i (not simplified).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form a + bi. Then state its real part and its imaginary part (each a real number).

a + bi: _____

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

$6 - 2\sqrt{5}\cdot i$; 6; $-2\sqrt{5}$

GRADING NOTE Form required: a + bi, then both parts. $\sqrt{(-20)} = 2\sqrt{5}\ i$. The imaginary part is the real number $-2\sqrt{5}$, not $-2\sqrt{5}\ i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write $(-4 + \sqrt{(-75)}) / 2$ in the form a + bi. Then state its real part and its imaginary part (each a real number).

a + bi: _____

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

$-2 + (5\sqrt{3}/2)i$; -2; $5\sqrt{3}/2$

GRADING NOTE Form required: a + bi, then both parts. $\sqrt{(-75)} = 5\sqrt{3}\, i$; divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write (10 + √(−64)) / 4 in the form a + bi. Then state its real part and its imaginary part (each a real number).

a + bi: _____

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

5/2 + 2i; 5/2; 2

GRADING NOTE Form required: a + bi, then both parts (5/2 or 2.5). √(−64) = 8i; 10/4 = 5/2 and 8/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5√2

GRADING NOTE Form required: simplest radical form. 50 = 25 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify √72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6√2

GRADING NOTE Form required: simplest radical form. 72 = 36 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 4√3

B. 3√2

C. 6√2

D. 2√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2√3

GRADING NOTE $12 = 4 \cdot 3$, so $\sqrt{12} = 2\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. A square root of a sum is not the sum of the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify √28.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2√7

GRADING NOTE Form required: simplest radical form. 28 = 4 · 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q5**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9 or x = −9

GRADING NOTE Any order. Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 = (-3)(-3) = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$?

A. No: only the number 0 has a square that is negative, and $0^2 = 0$, not -4.

B. Yes: $x = -2$ works, because the square of a negative number is negative: $(-2)^2 = -4$.

C. Yes: $x = 2i$ is a real number whose square is -4.

D. No: the square of every real number is 0 or positive, so it cannot equal -4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. No: the square of every real number is 0 or positive, so it cannot equal −4.

GRADING NOTE Squares of real numbers are never negative. The others square −2 incorrectly, call 2i a real number, or state a false fact ($0^2 = 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. 18

B. $\sqrt{-36}$

C. -6

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −6

GRADING NOTE $(-6)^2 = 36$. $\sqrt{-36}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

49

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write (8 + 6√2) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 3√2

GRADING NOTE Any equivalent form. Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (−6 + 4√5) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3 + 2√5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both terms must be divided: $(10 + \sqrt{7})/5 = 2 + \sqrt{7}/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

(12 − 9√3) / 3 equals

A. 4 − 3√3

B. 4 − 9√3

C. 4 − 3√1

D. 12 − 3√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 − 3√3

GRADING NOTE 12/3 = 4 and 9√3/3 = 3√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write (−4 + 2√3) / (−2) as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2 − √3

GRADING NOTE Any equivalent form. −4/(−2) = 2; 2√3/(−2) = −√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and (x − 3) + 2yi = 5 − 8i.
Find x and y.

x = _____

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8; −4

GRADING NOTE Form required: both values. Equal complex numbers have equal real parts (x − 3 = 5) and equal imaginary parts (2y = −8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and 3a + (b + 1)i = −12 + 4i. Find a and b.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−4; 3

GRADING NOTE Form required: both values. 3a = −12 and b + 1 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + yi = 7 + 0i, so x = 7 and y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and (p + q) + (p − q)i = 10 + 2i. What is p?

A. 6

B. 5

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE p + q = 10 and p − q = 2 give p = 6 and q = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

m = _____

The number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−3; 5

GRADING NOTE Form required: both. The imaginary part m + 3 must be 0, so m = −3 and $m^2 - 4 = 9 - 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for −3 + 5i.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 5)

GRADING NOTE Form required: an ordered pair (real part, imaginary part).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point (4, −2) in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 − 2i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{(6^2 + 8^2)} = 10$ (the modulus $|6 + 8i|$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i, the point (a, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. -7

B. $7 + i$

C. $-7i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID
S1-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. $-7i$

GRADING NOTE $-7i = 0 - 7i$, the point $(0, -7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7i

GRADING NOTE Form required: a + bi. 3 − 7 = −4 and 5i + 2i = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify (6 − 4i) − (−2 + 9i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13i

GRADING NOTE Form required: a + bi. Subtract both parts: 6 − (−2) = 8 and −4i − 9i = −13i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify (−1 + √(−16)) + (5 − √(−9)). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

4 + i

GRADING NOTE Form required: a + bi. (−1 + 4i) + (5 − 3i) = 4 + i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Form required: a + bi or the real number 4. Real: 2.5 + 1.5 = 4; imaginary: −1 − 3 + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which
(4 − 7i) + w = −2 + 3i.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

50

QUESTION ID
S2-M-Q5

TYPE
Short answer

CORRECT ANSWER
−6 + 10i

GRADING NOTE Form required: a + bi. w = (−2 + 3i) − (4 − 7i) = −6 + 10i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (3 + 5x) + (−7 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7x − 4

GRADING NOTE Any equivalent simplified form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (6 − 4x) − (−2 + 9x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−13x + 8

GRADING NOTE Any equivalent simplified form. The minus sign applies to both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

(2 − 3y) − (5 − y) equals

A. −3 − 2y

B. 7 − 4y

C. −3 + 2y

D. −3 − 4y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3 − 2y

GRADING NOTE 2 − 5 = −3 and −3y + y = −2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 + 4t.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2t - 6t = -4t$, so the sum is $5 - 4t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (0.5 − x) − (1.5 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−3x − 1

GRADING NOTE Any equivalent simplified form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE $-1 - 3 = -4$ and $-4 + 4 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2.5 + 1.5 = 4.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−16) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

64

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

65

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

√(−25) equals

A. −5

B. 25i

C. −5i

D. 5i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

66

QUESTION ID

S2-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5i

GRADING NOTE √(−25) = √25 · i = 5i (the principal root).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

67

QUESTION ID

S2-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of i as the principal square root of −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

68

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

k is a real number. For what value of k is (3 + ki) + (k − 5i) a real number? Give that real number.

k = _____

The sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

70

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5; 8

GRADING NOTE Form required: k and the number. The imaginary part k − 5 must be 0; then the sum is 3 + 5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
(2k − 6) + 4i − (k + 1 − i) a pure imaginary number
(real part 0)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

71

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. Real part: 2k − 6 − k − 1 = k − 7 = 0. The sum is then 5i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its conjugate is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

72

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a + bi) + (a − bi) = 2a, which is 0 only when a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and (a + 3i) − (2 + ai) is real. What is a?

A. 0

B. 2

C. 3

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

73

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Imaginary part 3 − a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

a and b are real numbers. Why is the sum of a + bi and a − bi always a real number?

A. The real parts a and a cancel, leaving 2bi, which is real.

B. The imaginary parts bi and −bi add to 0, leaving 2a, which is real.

C. Any sum of two complex numbers is real.

D. It is real only when b = 0, because otherwise the sum has an i term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

74

QUESTION ID

S2-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The imaginary parts bi and −bi add to 0, leaving 2a, which is real.

GRADING NOTE (a + bi) + (a − bi) = 2a + 0i. The others cancel the wrong parts, overgeneralize ((1 + i) + (1 + i) = 2 + 2i is not real), or forget that bi − bi = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

In the complex plane (real axis horizontal, imaginary axis vertical), the number a + bi is the point (a, b). Which point represents the sum (2 + i) + (1 + 3i)?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

76

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4)

GRADING NOTE Form required: the ordered pair. The sum is 3 + 4i, the point (3, 4): add the real parts and the imaginary parts, as with vectors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral in the complex plane. What kind of quadrilateral is it?

A. a parallelogram

B. a kite that is not a parallelogram

C. a trapezoid that is not a parallelogram

D. a square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

77

QUESTION ID

S2-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. a parallelogram

GRADING NOTE Adding complex numbers adds their vectors (parallelogram rule). Sides are √5 and √10, so it is not a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

78

QUESTION ID
S2-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE z − w is the displacement from w to z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points 5 + 2i and 1 − i in the complex plane?

A. 25

B. √13

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

79

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE |(5 + 2i) − (1 − i)| = |4 + 3i| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

80

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2 + i$

GRADING NOTE Form required: a + bi. The average $((6 - 2i) + (-2 + 4i))/2 = 2 + i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply (3 + 2i)(4 − 5i). Write the product in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

83

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 − 7i

GRADING NOTE Form required: a + bi. $12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

84

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 − 4i

GRADING NOTE Form required: a + bi. $4 - 4i + i^2 = 4 - 4i - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

85

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

34

GRADING NOTE Form required: a + bi or the real number 34. $25 + 15i - 15i - 9i^2 = 25 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

86

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + 2i

GRADING NOTE Form required: a + bi. $(1 + i)^2 = 2i$, and $2i(1 + i) = 2i + 2i^2 = -2 + 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

87

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 4i

GRADING NOTE Form required: a + bi. 23 = 4 · 5 + 3, so $i^{23} = i^3 = -i$; $-i(4 - i) = -4i + i^2 = -1 - 4i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply (3 + 2x)(4 − 5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

89

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12 - 7x - 10x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

90

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$4 - 4x + x^2$

GRADING NOTE Any equivalent form. Not $x^2 + 4$ (missing the middle term).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(5 − 3x)(5 + 3x) equals

A. $25 + 9x^2$

B. $10 − 9x^2$

C. $25 − 9x^2$

D. $25 − 30x − 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

91

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $25 − 9x^2$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

92

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(1 + x)^2 = 1 + 2x + x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply $x(4 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

93

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$4x - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

95

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. Definition of i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

96

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-i$

GRADING NOTE Any equivalent form. $i^3 = i^2 \cdot i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

97

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. $i^4 = (i^2)^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

98

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-10i^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^8

B. i^9

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

99

QUESTION ID

S3-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. i^8

GRADING NOTE Powers of i repeat every 4; 8 is a multiple of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

101

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 23 = 4 · 5 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

102

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 50 = 4 · 12 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

103

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE $36 = 4 \cdot 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

104

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $101 = 4 \cdot 25 + 1$, so the remainder is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

q = _____

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

105

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9; 3

GRADING NOTE Form required: both. 39 = 4 · 9 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is (a + 2i)(3 − i) a real number? Give the product.

a = _____

The product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

107

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6; 20

GRADING NOTE Form required: both. (a + 2i)(3 − i) = (3a + 2) + (6 − a)i; the imaginary part 6 − a = 0. Product 18 + 2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

108

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Product: (2 − b) + (2 + b)i; real part 2 − b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

109

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example (1 + i)(1 − i) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 0$

B. $c = 1$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

110

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. c = 0

GRADING NOTE $(c + i)^2 = c^2 - 1 + 2ci$; imaginary part $2c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

a and b are real numbers. Which work shows that $(a + bi)(a - bi)$ is always a real number?

A. $(a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 - b^2(-1) = a^2 + b^2$, which is real.

B. $(a + bi)(a - bi) = a^2 - b^2$, which is real.

C. $(3 + 2i)(3 - 2i) = 13$ is real, so every such product is real.

D. $(a + bi)(a - bi) = a^2 - b^2i^2$, and $a^2 - b^2i^2$ is real because it has no i in it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

111

QUESTION ID

S3-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A.

$(a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 - b^2(-1) = a^2 + b^2$, which is real.

GRADING NOTE The middle terms cancel and $i^2 = -1$. The others treat i^2 as having no i (without simplifying), lose the sign from $i^2 = -1$, or check one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

113

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11/5 − (2/5)i

GRADING NOTE Form required: a + bi. Multiply by (1 − 2i)/(1 − 2i): (3 + 4i)(1 − 2i)/5 = (11 − 2i)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write 10 / (3 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

114

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 + i

GRADING NOTE Form required: a + bi. 10(3 + i)/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 − 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

115

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the conjugate 2 + 5i; $(2 - 5i)^2$ is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. −1 + 2i

B. 2 + i

C. 1 − 2i

D. 1 + 2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

116

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 − 2i

GRADING NOTE (2 + i)(−i)/(i · −i) = (−2i + 1)/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

c and d are real and not both 0. Why does multiplying by the conjugate of the denominator c + di always give a real denominator?

A. $(c + di)(c - di) = c^2 - d^2$, a real number (the middle terms cancel).

B. The conjugate $c - di$ is itself a real number, so any product with it is real.

C. Multiplying by any complex number makes the denominator real.

D. $(c + di)(c - di) = c^2 + d^2$, a real number (positive because c and d are not both 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

117

QUESTION ID

S3-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $(c + di)(c - di) = c^2 + d^2$, a real number (positive because c and d are not both 0).

GRADING NOTE The product of a number and its conjugate is $c^2 + d^2$. The others lose the sign from $i^2 = -1$, call $c - di$ real, or overgeneralize $((1 + i)(1 + i) = 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-CN.C.7 • THE COMPLEX NUMBER SYSTEM

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

120

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7i or x = −7i

GRADING NOTE Form required: both solutions, exact. $x^2 = -49$, $x = \pm 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

121

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: both, a + bi. Discriminant 16 − 52 = −36; x = (4 ± 6i)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

122

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = −1/2 + (3/2)i or
x = −1/2 − (3/2)i

GRADING NOTE Form required: both, a + bi. Discriminant 4 − 40 = −36; x = (−2 ± 6i)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

123

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = -3 + 2\sqrt{5}\,i$ or
$x = -3 - 2\sqrt{5}\,i$

GRADING NOTE Form required: both, exact. $x + 3 = \pm\sqrt{-20} = \pm 2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

124

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 1 + (2\sqrt{3}/3)i$ or
$x = 1 - (2\sqrt{3}/3)i$

GRADING NOTE Form required: both, exact, a + bi. Discriminant $36 - 84 = -48 = (4\sqrt{3}\,i)^2$; $x = (6 \pm 4\sqrt{3}\,i)/6$. Check: product 7/3 and sum 2 agree with c/a and −b/a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

126

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 16 − 52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

127

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 4 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

128

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A negative discriminant gives two non-real solutions and no real ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

129

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3, −6, 7

GRADING NOTE Standard form $ax^2 + bx + c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Which is the quadratic formula for the solutions of $ax^2 + bx + c = 0$ $(a \neq 0)$?

A. $x = -b \pm \sqrt{(b^2 - 4ac)}/(2a)$

B. $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$

C. $x = (-b \pm \sqrt{(b^2 + 4ac)})/(2a)$

D. $x = (b \pm \sqrt{(b^2 - 4ac)})/(2a)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

130

QUESTION ID

S4-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$

GRADING NOTE The others change the sign of b, divide only the radical by 2a, or change the sign of 4ac.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−36) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

132

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−48) in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

133

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4√3 i

GRADING NOTE Form required: simplest radical form. 48 = 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

√(−20) equals

A. 2√5 i

B. 4√5 i

C. −2√5

D. 10i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

134

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2√5 i

GRADING NOTE √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

135

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-49)} = 7i.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write (−2 + √(−36)) / 4 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

136

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2 + (3/2)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 2)^2 = 9$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

138

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 or x = −1

GRADING NOTE Form required: both. $x - 2 = \pm 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

139

QUESTION ID

S4-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 + √5 or
x = −1 − √5

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

140

QUESTION ID

S4-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

141

QUESTION ID

S4-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 and −9

GRADING NOTE $x + 4 = \pm 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

142

QUESTION ID

S4-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 5\sqrt{2}$ or $x = -5\sqrt{2}$

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B3-Q5**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

b = _____

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

144

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−6; 13

GRADING NOTE Form required: both. With real coefficients the other solution is 3 + 2i; sum 6 = −b, product 9 + 4 = 13 = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Find b and c.

b = _____

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

145

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2; 17

GRADING NOTE Form required: both. The roots $-1 \pm 4i$ have sum $-2 = -b$ and product $1 + 16 = 17 = c$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

146

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Non-real solutions of real quadratics come in conjugate pairs (± in the formula).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

147

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 25 = 0$

GRADING NOTE $(5i)^2 + 25 = -25 + 25 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Why does $x^2 - (2 + i)x = 0$ have the solution $2 + i$ but not the solution $2 - i$?

A. Substituting $2 - i$ gives $(2 - i)^2 - (2 + i)(2 - i) = 0$, so $2 - i$ is a solution after all.

B. Non-real solutions of every quadratic equation come in conjugate pairs, so $2 - i$ must also be a solution, along with $2 + i$.

C. Its coefficient $2 + i$ is not real, so its solutions need not come in conjugate pairs; factoring $x(x - (2 + i)) = 0$ gives only 0 and $2 + i$.

D. A quadratic equation can have only one non-real solution, so once $2 + i$ is found there is no room for $2 - i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

148

QUESTION ID

S4-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Its coefficient 2 + i is not real, so its solutions need not come in conjugate pairs; factoring x(x − (2 + i)) = 0 gives only 0 and 2 + i.

GRADING NOTE The conjugate-pair rule needs real coefficients. The others apply the rule without that condition, misstate the number of solutions, or miscompute: $(2 - i)^2 - (2 + i)(2 - i) = 3 - 4i - 5 \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

150

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6 < k < 6$

GRADING NOTE Form required: an inequality. Discriminant $k^2 - 36 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

Consider $x^2 - 4x + c = 0$ with real c. (a) For which real values of c are the solutions non-real? (b) Solve the equation when c = 8.

(a) _____

(b) x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

151

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

c > 4; 2 ± 2i

GRADING NOTE Form required: both parts. The discriminant $16 - 4c < 0$ gives $c > 4$. For c = 8: $x = (4 \pm \sqrt{(-16)})/2 = 2 \pm 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

152

QUESTION ID

S4-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its discriminant $k^2 + 4$ is always positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when m =

A. 3

B. 36

C. −9

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

153

QUESTION ID

S4-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE $36 - 4m = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For $c > 4$, which gives the solutions of $x^2 - 4x + c = 0$ in terms of c?

A. $x = 2 \pm i\sqrt{(4 - c)}$

B. $x = -2 \pm i\sqrt{(c - 4)}$

C. $x = 2 \pm \sqrt{(c - 4)}$

D. $x = 2 \pm i\sqrt{(c - 4)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

154

QUESTION ID

S4-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $x = 2 \pm i\sqrt{(c - 4)}$

GRADING NOTE $(4 \pm \sqrt{(16 - 4c)})/2 = 2 \pm \sqrt{(4 - c)}$, and $4 - c < 0$, so $\sqrt{(4 - c)} = i\sqrt{(c - 4)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

157

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Any order of factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

158

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + i\sqrt{7})(x - i\sqrt{7})$

GRADING NOTE Any order. $x^2 - (i\sqrt{7})^2 = x^2 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into two linear factors with complex coefficients, each of the form $(2x + ki)$ with k real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

159

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(2x + 3i)(2x - 3i)$

GRADING NOTE Form required: factored. $4x^2 + 9 = (2x)^2 - (3i)^2$. The order of the factors does not matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

160

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE Any order. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

161

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - (3 + 2i))(x - (3 - 2i))$

GRADING NOTE Any equivalent form. Its zeros are $3 \pm 2i$ (discriminant -16). Check: sum 6, product $9 + 4 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

163

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7)(x − 7)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

164

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3x + 2)(3x − 2)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 - 1)(x^2 + 1)$

D. $(x^2 + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

165

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x^2 - 1)(x^2 + 1)$

GRADING NOTE $(x^2)^2 - 1^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

166

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

167

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + \sqrt{5})(x - \sqrt{5})$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

169

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25$

GRADING NOTE Any equivalent form. $-25i^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

170

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4x^2 + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

171

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3i)^2 = 9i^2 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 - 2x + 2$

C. $x^2 + 2$

D. $x^2 - 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

172

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 - 2x + 2$

GRADING NOTE Sum 2, product 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

173

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 25 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

175

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±5i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

Solve $x^2 - 6x + 13 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

176

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 3 \pm 2i$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

177

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Factor theorem, also for complex r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. 2i and −2i

B. 4i and −4i

C. −4

D. 2 and −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

178

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2i and −2i

GRADING NOTE $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 5 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

179

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = −1 ± 2i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)(x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$

GRADING NOTE Any order. $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, and $x^2 - 2x + 4$ has zeros $1 \pm i\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(x + i)(x − i)(x + 2i)(x − 2i)

GRADING NOTE Any order. $(x^2 + 1)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Discriminant $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - 3i$

D. $x - (-3/2 + (3\sqrt{3}/2)i)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D.

$x - (-3/2 + (3\sqrt{3}/2)i)$

GRADING NOTE $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$; $x^2 + 3x + 9$ has zeros $-3/2 \pm (3\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i. Give it in expanded form. Use ^ for exponents.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^4 - 2x^3 + 6x^2 - 8x + 8$

GRADING NOTE Conjugates −2i and 1 + i must also be zeros, so degree 4: $(x^2 + 4)(x^2 - 2x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. −4i and 2 − i are also zeros: 1 + 2 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Non-real zeros come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and −i. Its third zero is

A. −5

B. 5i

C. 1

D. i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. i

GRADING NOTE Conjugate of −i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Why must a polynomial with real coefficients and degree 3 have at least one real zero?

A. The Fundamental Theorem of Algebra says that every zero of a polynomial with real coefficients is real, so all 3 zeros are real.

B. It has 3 zeros counted with multiplicity, and its non-real zeros come in conjugate pairs, so at most 2 are non-real and at least 1 is real.

C. Every polynomial has at least one real zero, whatever its degree.

D. Every cubic with real coefficients can be factored over the integers, so at least one of its 3 zeros is a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. It has 3 zeros counted with multiplicity, and its non-real zeros come in conjugate pairs, so at most 2 are non-real and at least 1 is real.

GRADING NOTE Count with the Fundamental Theorem of Algebra and pair the non-real zeros. The others make false claims ($x^2 + 1$ has no real zero; the theorem guarantees complex zeros; $x^3 - 2$ has no integer zero).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? (Use the Fundamental Theorem of Algebra.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

194

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Form required: the number. A polynomial of degree n ≥ 1 has n complex zeros counted with multiplicity; the degree is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

$p(x) = (x - 2)^3(x^2 + 9)$. List every complex zero with its multiplicity.

Zero with multiplicity 3: _____

Zeros with multiplicity 1 (both): _____

Sum of the multiplicities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

195

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

2; x = 3i or x = −3i; 5

GRADING NOTE Form required: all three parts. $x^2 + 9 = (x - 3i)(x + 3i)$. 3 + 1 + 1 = 5, the degree, as the Fundamental Theorem of Algebra requires.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Which argument shows that every quadratic $p(x) = ax^2 + bx + c$ (real a, b, c, $a \neq 0$) has exactly two complex zeros, counted with multiplicity?

A. $x^2 - 5x + 6 = (x - 2)(x - 3)$ has two zeros, so every quadratic has two zeros.

B. By the quadratic formula the zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. If $b^2 - 4ac < 0$ the square root does not exist, so there are no zeros; otherwise there are two, so a quadratic never has more than two zeros.

C. The graph of every quadratic is a parabola, and a parabola that opens up or down always crosses the x-axis in exactly two places, so every quadratic $ax^2 + bx + c$ has exactly two zeros.

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

GRADING NOTE Every case of the discriminant gives two zeros counted with multiplicity (for $D < 0$, $\sqrt{D} = i\sqrt{(-D)}$ is a complex number). The others check one example, use a false graph fact ($x^2 + 1$ never crosses), or ignore complex zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$. How many are there, and does the count agree with the Fundamental Theorem of Algebra for degree 4?

All zeros: x = _____

Number of zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 or −3 or 3i or −3i; 4

GRADING NOTE Form required: all four zeros and the count. $x^4 - 81 = (x^2 - 9)(x^2 + 9) = (x - 3)(x + 3)(x - 3i)(x + 3i)$; 4 = the degree.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Which statement is correct?

A. The student is right: a polynomial that does not factor over the integers has no zeros.

B. It has exactly 3 real zeros, because the Fundamental Theorem of Algebra says a polynomial of degree 3 has 3 zeros, and a polynomial with real coefficients has only real zeros.

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

D. It has 3 complex zeros counted with multiplicity, and all three could be non-real, because a polynomial that does not factor over the integers need not cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

GRADING NOTE Degree 3 gives 3 zeros (Fundamental Theorem of Algebra); conjugate pairs make the number of non-real zeros even, so at least 1 is real. The others confuse "no integer factor" with "no zero", assume all zeros are real, or ignore conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 4

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 1 + 1 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The highest power is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

Find the zeros of $(x - 2)(x + 5)$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 or −5

GRADING NOTE Form required: both zeros, in any order. Each factor equals 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

$p(x) = (x - 1)^3$. Find the zero of p and its multiplicity.

Zero: _____

Multiplicity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1; 3

GRADING NOTE Form required: both. The factor $x - 1$ appears three times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Zero-product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of x(x + 4) are

A. 0 and 4

B. 0 and −4

C. −4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 and −4

GRADING NOTE x = 0 or x + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

Find the zeros of (2x − 3)(x + 1).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/2 or −1

GRADING NOTE Form required: both zeros, in any order (3/2 or 1.5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 9 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

212

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±3i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 2 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

213

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 ± i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

214

QUESTION ID

S6-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ±i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $1 \pm 2i$

D. $2 \pm i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

215

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 ± 2i

GRADING NOTE Discriminant −16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = -81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

216

QUESTION ID

S6-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = ±9i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

218

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 zeros in all, 2 real (the crossings), so 2 non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

$p(x) = x^3 - 4x^2 + 5x - 2 = (x - 1)^2(x - 2)$. Give the multiplicity of each zero.

Multiplicity of the zero 1: _____

Multiplicity of the zero 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

219

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2; 1

GRADING NOTE Form required: both. The factor $(x - 1)$ appears twice and $(x - 2)$ once; $2 + 1 = 3$ = degree. The graph touches the x-axis at 1 and crosses at 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

220

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 6 zeros are non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 4

B. 5

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

221

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $5 - 1 = 4$ (two conjugate pairs).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1, and no other real zeros?

A. No: the fourth zero (counted with multiplicity) would be a single non-real zero, but non-real zeros come in conjugate pairs.

B. No: a degree-4 polynomial with real coefficients must always have exactly four distinct real zeros.

C. Yes: a degree-4 polynomial can have any number of zeros from 0 to 4.

D. Yes: for example $(x - 1)(x - 2)(x - 3)(x - i)$ has degree 4, real coefficients, and exactly three real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

222

QUESTION ID

S6-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. No: the fourth zero (counted with multiplicity) would be a single non-real zero, but non-real zeros come in conjugate pairs.

GRADING NOTE 3 single real zeros leave one zero, which must be real (no partner). The others use a polynomial with a non-real coefficient, ignore conjugate pairs, or ignore non-real zeros and repeated zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Which argument proves that every polynomial of odd degree with real coefficients has at least one real zero?

A. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come one at a time, so there is an odd number of them, and an odd number of non-real zeros leaves at least one zero that is real.

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

C. The Fundamental Theorem of Algebra says that every polynomial of degree $n \geq 1$ with real coefficients has at least one real zero; a polynomial of odd degree has degree at least 1, so it has at least one real zero as well.

D. $x^3 - 1$ has the real zero 1, so every polynomial of odd degree has a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

224

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

GRADING NOTE Count zeros with the theorem and pair the non-real ones. The others check one example, misstate the theorem (it guarantees complex zeros, and $x^2 + 1$ has no real zero), or claim an odd number of non-real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. Which describes all of its zeros, counted with multiplicity?

A. 1 + i, 1 − i, and 3 only (a degree-4 polynomial has 3 zeros here).

B. 1 + i, 3, and two more non-real zeros that need not include 1 − i.

C. 1 + i, 1 − i, 3, and one more real zero r (r may equal 3, making 3 a double zero).

D. 1 + i, 1 − i, 3, and −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

225

QUESTION ID
S6-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. 1 + i, 1 − i, 3, and one more real zero r (r may equal 3, making 3 a double zero).

GRADING NOTE Degree 4 gives four zeros; 1 − i must also be a zero, and the fourth zero must be real (a non-real one would need a partner). The others count only three, omit the conjugate, or assume the fourth zero is −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients must have at least one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

226

QUESTION ID
S6-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x^4 + 1$ has no real zeros; only odd degree guarantees a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 1, 3, 5, 7

B. 7 only

C. 0, 2, 4, 6

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

227

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1, 3, 5, 7

GRADING NOTE The number of real zeros is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Which statement about the Fundamental Theorem of Algebra is correct?

A. It guarantees that a polynomial of degree n ≥ 1 has n complex zeros counted with multiplicity, but gives no method to find them.

B. It says that every zero of a polynomial with real coefficients is real, so a polynomial of degree n has n real zeros counted with multiplicity.

C. It gives a formula, like the quadratic formula, that finds the exact zeros of every polynomial of any degree.

D. It applies only to polynomials that factor over the integers, so it says nothing about other polynomials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

228

QUESTION ID

S6-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. It guarantees that a polynomial of degree n ≥ 1 has n complex zeros counted with multiplicity, but gives no method to find them.

GRADING NOTE The theorem is an existence and counting statement. The others claim a formula (none exists in general for degree 5 or more), claim real zeros ($x^2 + 1$), or add a false condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a • SEEING STRUCTURE IN EXPRESSIONS

Interpret the parts of polynomial, rational, and exponential expressions in context

MAIN	Algebra II main question	**HSA-SSE.A.1.a**
BACKWARD 1	Name terms, factors, and coefficients	**HSA-SSE.A.1.a**
BACKWARD 2	Translate between a percent change and a multiplying factor	**7.RP.A.3**
BACKWARD 3	Interpret a quotient as an amount per unit	**6.RP.A.2**
FORWARD 1	Build an expression whose parts have given meanings	**HSA-SSE.A.1.a**
FORWARD 2	Rewrite an exponential expression to reveal a different rate	**HSF-IF.C.8.b**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A town's population is modeled by $P(t) = 12{,}500(0.97)^t$, where t is the number of years after 2020. Which interpretation of 12,500, 0.97, and $1 - 0.97$ is correct?

A. 12,500 is the population in 2020; 0.97 means the population decreases by 97% per year; $1 - 0.97 = 0.03$ is the population left after one year.

B. 12,500 is the population in 2020; 0.97 is the yearly factor (each year the population is 97% of the year before); $1 - 0.97 = 0.03$ means a 3% decrease per year.

C. 12,500 is the number of people lost per year; 0.97 is the yearly factor; $1 - 0.97 = 0.03$ means a 3% decrease per year.

D. 12,500 is the population after one year; 0.97 is the yearly factor; $1 - 0.97 = 0.03$ means a 3% increase per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

231

QUESTION ID

S7-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 12,500 is the population in 2020; 0.97 is the yearly factor (each year the population is 97% of the year before); $1 - 0.97 = 0.03$ means a 3% decrease per year.

GRADING NOTE At $t = 0$, $P = 12{,}500$; multiplying by 0.97 keeps 97%, a 3% decrease. The others read 0.97 as the percent lost, call 12,500 the value after one year or the yearly loss, or read 0.03 as growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The average cost per item of making x items is C(x) = (4,000 + 6x)/x dollars. Which interpretation of 4,000, 6x, and the division by x is correct?

A. 4,000 is a fixed cost in dollars; 6x is the variable cost; dividing by x gives the total cost of x items.

B. 4,000 is the cost of each item; 6x is the number of items made; dividing by x gives the total cost of all items.

C. 4,000 is a fixed cost in dollars; 6x is the variable cost ($6 per item for x items); dividing the total cost by x gives the cost per item.

D. 4,000 is a fixed cost in dollars; 6x is the profit on x items ($6 per item); dividing by x gives the profit per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

232

QUESTION ID

S7-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 4,000 is a fixed cost in dollars; 6x is the variable cost ($6 per item for x items); dividing the total cost by x gives the cost per item.

GRADING NOTE Total cost = fixed + variable, and total ÷ items = average cost per item. The others misread one part: the fixed cost, the variable cost, or what the division gives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

An open box is made by cutting squares of side x cm from the corners of a 20 cm by 20 cm sheet and folding up the sides. Its volume is $V(x) = x(20 - 2x)^2$ cm³. Which interpretation of x, $20 - 2x$, and $(20 - 2x)^2$ is correct?

A. x is the side of the base; $20 - 2x$ is the height; $(20 - 2x)^2$ is the volume in cm³.

B. x is the height; $20 - 2x$ is the side of the base (20 cm minus one cut); $(20 - 2x)^2$ is the perimeter of the base.

C. x is the height; $20 - 2x$ is the area of one side; $(20 - 2x)^2$ is the surface area of the box.

D. x is the height of the box; $20 - 2x$ is the side length of the square base (20 cm minus two cuts); $(20 - 2x)^2$ is the base area in cm².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

233

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x is the height of the box; $20 - 2x$ is the side length of the square base (20 cm minus two cuts); $(20 - 2x)^2$ is the base area in cm².

GRADING NOTE The folded flaps have height x; each side of the base loses x at both ends. The others swap height and base, count one cut, or confuse area with perimeter or surface area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

The amount of a medicine in the body t hours after a dose is $A(t) = 200(1/2)^{t/6}$ mg. Which interpretation of 200, 1/2, and t/6 is correct?

A. 200 mg is the dose at t = 0; 1/2 means the amount is halved repeatedly; t/6 counts 6-hour periods, so the amount halves every 6 hours.

B. 200 mg is the dose at t = 0; 1/2 means the amount is halved; t/6 means the amount halves every 1/6 hour.

C. 200 mg is the amount left after 6 hours; 1/2 means the amount is halved; t/6 counts 6-hour periods.

D. 200 mg is the dose at t = 0; 1/2 means half of the dose is removed every hour; t/6 is the number of doses taken.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

234

QUESTION ID

S7-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 200 mg is the dose at t = 0; 1/2 means the amount is halved repeatedly; t/6 counts 6-hour periods, so the amount halves every 6 hours.

GRADING NOTE At t = 0, A = 200; the exponent t/6 increases by 1 every 6 hours, and each step multiplies by 1/2 (half-life 6 hours). The others misread the exponent, the starting value, or the halving period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

Maya keeps $1,500 in an account that earns 4% interest compounded once a year and also puts $200 per year in a jar (no interest). Her total after n years is $T(n) = 1500(1.04)^n + 200n$. Which interpretation of $1500(1.04)^n$, 1.04, and 200n is correct?

A. $1500(1.04)^n$ is the account balance after n years; 1.04 means 104% interest per year; 200n is the cash in the jar after n years.

B. $1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor (4% interest added each year); 200n is the cash in the jar after n years.

C. $1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor; 200n is the interest earned by the jar.

D. $1500(1.04)^n$ is the interest earned after n years; 1.04 is the yearly growth factor; 200n is the cash in the jar after n years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

235

QUESTION ID

S7-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. $1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor (4% interest added each year); 200n is the cash in the jar after n years.

GRADING NOTE The exponential term is the whole balance; 1.04 keeps 100% and adds 4%; 200n is the uninvested cash. The others call the balance interest, read 1.04 as 104% interest, or give the jar interest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Name terms, factors, and coefficients • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Which lists the terms of 4,000 + 6x?

A. 4,000 and 6

B. 4,000, 6, and x

C. 4,000 and 6x

D. 6 and x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

237

QUESTION ID

S7-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 4,000 and 6x

GRADING NOTE Terms are the parts that are added. The others list the factors 6 and x of the term 6x, or mix terms and factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of x in 7x − 3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

238

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

In 5x(x − 2), which is a factor?

A. 5x − 2

B. x − 10

C. x − 2

D. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

239

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x − 2

GRADING NOTE 5x(x − 2) is the product of 5, x and x − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 3 + 2x, the number 3 is a factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

240

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 is a term (it is added).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

Which lists the factors of $200(0.5)^t$?

A. 0.5 and t

B. $200 + (0.5)^t$

C. 200 and 0.5t

D. 200 and $(0.5)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

241

QUESTION ID

S7-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 200 and $(0.5)^t$

GRADING NOTE $200(0.5)^t$ is 200 times $(0.5)^t$. The others rewrite the power as a product, add instead of multiplying, or split the power into its base and exponent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Translate between a percent change and a multiplying factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A price is multiplied by 0.97. By what percent does it change? Give a signed percent (negative for a decrease).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

243

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−3%

GRADING NOTE 0.97 = 1 − 0.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What factor multiplies an amount that increases by 4%?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

244

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.04

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

Multiplying by 1.25 is the same as

A. increasing by 2.5%

B. increasing by 25%

C. increasing by 125%

D. decreasing by 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

245

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. increasing by 25%

GRADING NOTE 1 + 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 0.8 decreases an amount by 80%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

246

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 20%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

An amount is multiplied by 0.85. By what percent does it decrease? (Hint: find 1 − 0.85.)

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

247

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

15%

GRADING NOTE Any equivalent form (15% or 15). 1 − 0.85 = 0.15, so the amount decreases by 15%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 3

Interpret a quotient as an amount per unit • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A total cost of $600 is shared equally by 30 people. What does 600/30 = 20 represent?

A. The total cost, $20

B. The number of people per dollar

C. The cost per person, $20

D. The number of people, 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

249

QUESTION ID

S7-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The cost per person, $20

GRADING NOTE Dollars divided by people gives dollars per person. The others invert the rate or misread the quotient as a total or a count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

A car travels 150 miles in 3 hours. What does 150/3 = 50 represent?

A. The average speed, 50 miles per hour

B. The time per mile, 50 hours per mile

C. The total distance, 50 miles

D. The number of hours, 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

250

QUESTION ID

S7-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The average speed, 50 miles per hour

GRADING NOTE Miles divided by hours gives miles per hour. The others invert the rate or read the quotient as a total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

If T is a total cost for n items, then T/n is

A. the profit

B. the number of items

C. the cost per item

D. the total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

251

QUESTION ID

S7-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the cost per item

GRADING NOTE Unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Dividing a total by the number of items gives an average per item.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

252

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a mean or unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

Making x items costs a total of 4,000 + 6x dollars. What does (4,000 + 6x)/x equal when x = 100, and what does it mean?

A. $46: the cost of making the 100th item alone

B. $40: the fixed cost per item when 100 items are made

C. $4,600: the total cost of making 100 items

D. $46: the average cost per item when 100 items are made

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

253

QUESTION ID

S7-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $46: the average cost per item when 100 items are made

GRADING NOTE 4,600/100 = 46 dollars per item, the average. The others give the total, confuse the average with the cost of one more item ($6), or include only the fixed cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Build an expression whose parts have given meanings • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A $24,000 car loses 15% of its value each year. Write an expression for its value in dollars t years after purchase.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

255

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$24000(0.85)^t$

GRADING NOTE Any equivalent expression. Each year the car keeps 85% of its value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $90 joining fee and $35 per month. Write an expression for the average cost per month, in dollars, over m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

256

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(90 + 35m)/m

GRADING NOTE Any equivalent expression is accepted by the checker (for example 35 + 90/m).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

True or false?

An expression for a quantity that triples every 5 years, starting at 40, is $40(3)^{5t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

257

QUESTION ID
S7-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The exponent counts 5-year periods: $40(3)^{t/5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bacteria count starts at 500 and doubles every 3 hours. Which expression gives the count after h hours?

A. $2(500)^{h/3}$

B. $500(2)^{3h}$

C. $500(3)^{h/2}$

D. $500(2)^{h/3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

258

QUESTION ID

S7-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $500(2)^{h/3}$

GRADING NOTE h/3 doubling periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

A gym charges a $90 joining fee and $35 per month, so the average cost per month over m months is 35 + 90/m dollars. What does the part 90/m represent?

A. The monthly fee, which stays the same $35 for every m.

B. The joining fee spread over the m months; it shrinks as m grows.

C. The total amount paid for the gym after m months of membership.

D. The joining fee, charged again in full each month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

259

QUESTION ID
S7-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. The joining fee spread over the m months; it shrinks as m grows.

GRADING NOTE 90/m is the share of the one-time $90 fee per month. The others confuse it with the $35 term, the total 90 + 35m, or a repeated fee (which would be 90, not 90/m).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Rewrite an exponential expression to reveal a different rate • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $12{,}500(0.97)^t$ (t in years) to show the monthly factor, rounded to 5 decimal places.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

261

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$12{,}500(0.99746)^{12t}$

GRADING NOTE Rounding to 5 decimal places required for the factor. $0.97^{1/12} \approx 0.997465$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

$A(t) = 200(1/2)^{t/6}$. Write A as $200 \cdot b^t$. Give b to 4 decimal places, and the hourly percent decrease to the nearest hundredth of a percent.

b = _____

Hourly percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

262

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8909; 10.91%

GRADING NOTE Rounding as stated. $b = (1/2)^{1/6} \approx 0.890899$, and $1 - 0.890899 \approx 0.1091$, a 10.91% decrease per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

True or false?

$1.04^{n} = (1.04^{1/12})^{12n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

263

QUESTION ID

S7-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power: (1/12)(12n) = n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

Which expression shows that $500(2)^{t/3}$ grows by about 26% per time unit?

A. $1.26(500)^t$

B. $500(1.26)^{3t}$

C. $500(1.26)^t$

D. $500(2.26)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

264

QUESTION ID

S7-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $500(1.26)^t$

GRADING NOTE $2^{1/3} \approx 1.26$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

$A(t) = 200(1/2)^{t/6}$ is rewritten as $A(t) = 200(0.8909)^t$ (rounded). Why does an exact rewriting such as $200((1/2)^{1/6})^t$ give the same value as $200(1/2)^{t/6}$ for every t?

A. By the power of a power rule, $((1/2)^{1/6})^t = (1/2)^{t/6}$, so the two are equal for every t; only the displayed rate differs.

B. Rewriting an exponential always changes its values a little, so the two expressions are only close for each t.

C. They give the same value only at t = 0, where both equal 200; for other t the base 0.8909 makes them differ.

D. They are equal only when t is a multiple of 6, because the exponent t/6 is then a whole number and no rounding is involved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

265

QUESTION ID

S7-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. By the power of a power rule, $((1/2)^{1/6})^t = (1/2)^{t/6}$, so the two are equal for every t; only the displayed rate differs.

GRADING NOTE $(a^m)^n = a^{mn}$ makes the forms identical. The others claim agreement only at some inputs, or confuse the exact rewriting with the rounded factor 0.8909.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b • SEEING STRUCTURE IN EXPRESSIONS

Interpret complicated expressions by viewing parts as single entities

MAIN	Algebra II main question	**HSA-SSE.A.1.b**
BACKWARD 1	Squares are never negative	**8.EE.A.2**
BACKWARD 2	Powers of numbers greater than 1 and between 0 and 1	**8.EE.A.1**
BACKWARD 3	A fraction gets smaller as its denominator grows	**3.NF.A.3.d**
FORWARD 1	Find the set of all values of an expression on a domain	**HSA-SSE.A.1.b**
FORWARD 2	Use single-entity reasoning to find horizontal asymptotes	**HSF-IF.C.7.d**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

$P(r) = 5000(1 + r)^{10}$ with $r > 0$ is the value of an investment. Which reasoning shows, without calculating, that $P(r) > 5000$?

A. $(1 + r)^{10} = 1 + r^{10}$ by distributing the exponent, and $1 + r^{10}$ is greater than 1 for $r > 0$, so $P(r) > 5000$.

B. $1 + r > 1$, so $(1 + r)^{10}$ is a product of ten numbers greater than 1 and is greater than 1; 5000 times a number greater than 1 is greater than 5000.

C. For $r = 0.05$, $P = 8{,}144.47 > 5000$, so $P(r) > 5000$ for every $r > 0$.

D. The exponent 10 is greater than 1, so any positive number raised to the 10th power is greater than 1, and $P(r) > 5000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

268

QUESTION ID

S8-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $1 + r > 1$, so $(1 + r)^{10}$ is a product of ten numbers greater than 1 and is greater than 1; 5000 times a number greater than 1 is greater than 5000.

GRADING NOTE Treat $(1 + r)^{10}$ as one quantity greater than 1. The others distribute a power over a sum, check one value, or use a false rule ($0.5^{10} < 1$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

The height of a ball is $h(t) = 80 - 16(t - 2)^2$ feet after t seconds. Without expanding, use the fact that $(t - 2)^2 \geq 0$ to find the greatest height and when it occurs.

Greatest height (ft): _____

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

269

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

80; 2

GRADING NOTE Form required: both. $16(t - 2)^2 \geq 0$, so $h(t) \leq 80$, with equality only when $t - 2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

$C(x) = 10 + 40/(x + 1)$ is a cost in dollars for $x \geq 0$. Which reasoning shows that $C(x)$ is always more than 10 and never more than 50?

A. $40/(x + 1)$ is at least 40 because $x + 1 \geq 1$, so $C(x)$ is at least $10 + 40 = 50$ for every $x \geq 0$.

B. $40/(x + 1)$ can be 0, so $C(x)$ can equal 10; it is at most 50 because $10 + 40 = 50$.

C. $C(0) = 50$ and $C(100) \approx 10.4$, and both are between 10 and 50, so every value of $C(x)$ is between 10 and 50.

D. $x + 1 \geq 1$, so $40/(x + 1)$ is positive and at most 40; adding 10 gives more than 10 and at most 50 (equal to 50 at $x = 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

270

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x + 1 \geq 1$, so $40/(x + 1)$ is positive and at most 40; adding 10 gives more than 10 and at most 50 (equal to 50 at $x = 0$).

GRADING NOTE Treat $40/(x + 1)$ as one positive quantity at most 40. The others check two values, reverse the inequality, or claim the fraction can be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

The temperature of a drink is $T(t) = 70 - 30(0.8)^t$ °F after t minutes ($t \geq 0$). Which reasoning explains why T(t) is always below 70 and gets closer to 70 as t grows?

A. $30(0.8)^t$ is always positive, so $T < 70$; $(0.8)^t$ shrinks toward 0 as t grows, so T approaches 70.

B. $(0.8)^t$ grows as t grows, so the amount $30(0.8)^t$ grows, and T rises toward 70 from below without reaching it.

C. $30(0.8)^t$ becomes exactly 0 after enough minutes, and from then on T equals 70 and stops changing.

D. $T(0) = 40$ and $T(10) \approx 66.8$ are both below 70 and rising, so T stays below 70 and approaches it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

271

QUESTION ID

S8-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $30(0.8)^t$ is always positive, so $T < 70$; $(0.8)^t$ shrinks toward 0 as t grows, so T approaches 70.

GRADING NOTE Treat $30(0.8)^t$ as one positive, shrinking quantity. The others claim the power grows, claim it reaches 0, or reason from two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

$f(x) = 3/(x^2 + 1)$. Which gives the greatest value of f(x) with a correct reason?

A. 4, at x = 1, because $x^2 + 1$ is smallest when x = 1.

B. 3, at x = 0: $x^2 + 1 \geq 1$, so $3/(x^2 + 1) \leq 3$, with equality only when x = 0.

C. 3, at x = 0, because f(1) = 1.5 and f(2) = 0.6 are smaller.

D. There is no greatest value, because $x^2 + 1$ can be as large as we like.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

272

QUESTION ID

S8-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 3, at x = 0: $x^2 + 1 \geq 1$, so $3/(x^2 + 1) \leq 3$, with equality only when x = 0.

GRADING NOTE The smallest denominator gives the largest value. The others check a few inputs only, confuse a large denominator with a large value, or misplace the smallest denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Squares are never negative • Grade 8 • Expressions & Equations • 8.EE.A.2

True or false?

$(t - 2)^2 \geq 0$ for every real t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

274

QUESTION ID

S8-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A square of a real number is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the smallest value of $(x + 5)^2$, and for which x does it occur?

Smallest value: _____

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

275

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0; −5

GRADING NOTE Form required: both. A square is never negative and equals 0 only when x + 5 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

Which expression can be negative for some real x?

A. $x^2 + 1$

B. $x - 1$

C. x^2

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

276

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x − 1

GRADING NOTE $x - 1 < 0$ for $x < 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

What is the smallest value of $x^2 + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

277

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$-(x - 3)^2$ is never positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

278

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the opposite of a nonnegative number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Powers of numbers greater than 1 and between 0 and 1 • Grade 8 • Expressions & Equations • 8.EE.A.1

Is 1.05^{10} greater than or less than 1?

A. Less than 1

B. Equal to 1

C. Greater than 1

D. It cannot be decided without a calculator

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

280

QUESTION ID

S8-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Greater than 1

GRADING NOTE A power of a number greater than 1 (with a positive whole-number exponent) is greater than 1 (about 1.63).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

Is 0.8^5 greater than or less than 1?

A. It cannot be decided without a calculator

B. Greater than 1

C. Less than 1

D. Equal to 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

281

QUESTION ID

S8-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Less than 1

GRADING NOTE Each factor 0.8 makes the product smaller: $0.8^5 \approx 0.33$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As n increases, 0.5^n

A. becomes negative

B. stays 0.5

C. gets closer to 0

D. grows without bound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

282

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. gets closer to 0

GRADING NOTE Repeated halving.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

True or false?

A negative number raised to any whole-number power is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

283

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-2)^2 = 4$ is positive; even powers of negatives are positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

Which order is from least to greatest?

A. 0.9^5, 0.9^2, 0.9

B. 0.9^2, 0.9^5, 0.9

C. 0.9^2, 0.9, 0.9^5

D. 0.9, 0.9^2, 0.9^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

284

QUESTION ID

S8-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 0.9^5, 0.9^2, 0.9

GRADING NOTE For $0 < b < 1$, higher powers are smaller: $0.9^5 \approx 0.59 < 0.9^2 = 0.81 < 0.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 3

A fraction gets smaller as its denominator grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, 40/2 or 40/5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

286

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

40/2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

True or false?

If $x \geq 0$, then $40/(x + 1) \leq 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

287

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The denominator is at least 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows, 6/(x + 3)

A. increases

B. becomes negative

C. decreases toward 0

D. stays 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

288

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. decreases toward 0

GRADING NOTE The denominator grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

What is the greatest value of 12/(x + 4) for x ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

289

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

True or false?

8/10 is greater than 8/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

290

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Same numerator, larger denominator gives a smaller fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Find the set of all values of an expression on a domain • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

Find every value that $h(t) = 80 - 16(t - 2)^2$ takes for $0 \le t \le 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

292

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$16 \le h \le 80$

GRADING NOTE Form required: an inequality or interval. $(t - 2)^2$ goes from 0 to 4, so $16(t - 2)^2$ from 0 to 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

Find every value of $f(x) = 3/(x^2 + 1)$ for real x. Write the answer as a compound inequality in y.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

293

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 < y \leq 3$

GRADING NOTE Form required: the compound inequality. $x^2 + 1 \geq 1$, so $0 < 3/(x^2 + 1) \leq 3$, with 3 at $x = 0$ and values near 0 for large $|x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

True or false?

For $x \geq 0$, $C(x) = 10 + 40/(x + 1)$ takes every value in $10 < C \leq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

294

QUESTION ID

S8-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 40/(x + 1) takes every value in (0, 40].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

For $t \geq 0$, which set contains every value of $70 - 30(0.8)^t$?

A. $T \geq 70$

B. $40 \leq T < 70$

C. $40 < T \leq 70$

D. $0 < T < 70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

295

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $40 \leq T < 70$

GRADING NOTE $30(0.8)^t$ goes from 30 (at $t = 0$) toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

Find every value of $g(x) = 5 - 2/(x^2 + 4)$ for real x.
Write the answer as a compound inequality in y.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

296

QUESTION ID
S8-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$4.5 \leq y < 5$

GRADING NOTE Form required: the compound inequality. $2/(x^2 + 4)$ is greater than 0 and at most 0.5 (at $x = 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Use single-entity reasoning to find horizontal asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Why does y = 10 + 40/(x + 1) have the horizontal asymptote y = 10?

A. The constant term of any rational expression, here 10, is always its horizontal asymptote.

B. As x grows without bound, 40/(x + 1) gets closer and closer to 0, so y gets closer to 10.

C. When x = −1 the fraction is undefined, so y = 10 there.

D. At x = 0, y = 10 + 40 = 50, and 50 is the asymptote shifted up by the 40 in the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

298

QUESTION ID

S8-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. As x grows without bound, 40/(x + 1) gets closer and closer to 0, so y gets closer to 10.

GRADING NOTE The quotient 40/(x + 1) tends to 0. The others use a single point, a false rule (y = 10 + 40x/(x + 1) has asymptote 50), or confuse the vertical asymptote x = −1 with a value of y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

Rewrite (2x + 7)/(x + 3) as 2 + 1/(x + 3) and state its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

299

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2

GRADING NOTE Form required: the equation y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

True or false?

$y = 3/(x^2 + 1)$ has the horizontal asymptote $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

300

QUESTION ID

S8-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3/(x^2 + 1)$ approaches 0 as $|x|$ grows; the asymptote is $y = 0$ (3 is the maximum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

The horizontal asymptote of $y = 5 - 2/(x - 4)$ is

A. $y = -2$

B. $y = 5$

C. $y = 4$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

301

QUESTION ID
S8-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. y = 5

GRADING NOTE $-2/(x - 4)$ approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

Does the graph of y = 10 + 40/(x + 1) ever reach its horizontal asymptote y = 10?

A. Yes: for very large x the fraction is 0, so y equals 10.

B. No: the graph stays far from y = 10 for every x.

C. Yes: at x = 0, where y = 10.

D. No: 40/(x + 1) is never 0, so y is never exactly 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

302

QUESTION ID

S8-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. No: 40/(x + 1) is never 0, so y is never exactly 10.

GRADING NOTE A fraction with numerator 40 is never 0. The others compute y(0) incorrectly (it is 50), treat "close" as "equal," or deny that y approaches 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

Factor higher-degree expressions by recognizing differences of squares and sums or differences of cubes

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Take out a greatest common factor	**6.EE.A.3**
BACKWARD 2	Recognize perfect squares and perfect cubes	**8.EE.A.2**
BACKWARD 3	Multiply to check a factoring pattern	**HSA-APR.A.1**
FORWARD 1	Factor expressions whose parts are themselves expressions	**HSA-SSE.A.2**
FORWARD 2	Complete the factorization over the complex numbers	**HSN-CN.C.8**

SET 9 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 81$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

305

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 9)(x + 3)(x - 3)$

GRADING NOTE Form required: factored completely over the integers. $(x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9)$. $x^2 + 9$ does not factor over the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

Factor $8x^3 + 27$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

306

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 3)(4x^2 - 6x + 9)$

GRADING NOTE Form required: factored completely. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 2x$, $b = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

Factor $x^6 - 64$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

307

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4)$

GRADING NOTE Form required: factored completely, any order. $(x^3 - 8)(x^3 + 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

Factor $16x^4 - y^4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

308

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(4x^2 + y^2)(2x + y)(2x - y)$

GRADING NOTE Form required: factored completely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

Factor $2x^5 - 32x$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

309

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x^2 + 4)(x + 2)(x - 2)$

GRADING NOTE Form required: factored completely. Common factor 2x first: $2x(x^4 - 16)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 1

Take out a greatest common factor • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Factor 6x + 18.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

311

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

Factor $2x^5 - 32x$ by taking out the greatest common factor only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

312

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$2x(x^4 - 16)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

The greatest common factor of $12x^3$ and $18x$ is

A. $6x^3$

B. $6x$

C. $3x$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

313

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x

GRADING NOTE 6 and x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$3x^2 + 9x = 3x(x + 9)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

314

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3x(x + 9) = 3x^2 + 27x$; the correct factorization is $3x(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

Factor $5y^2 - 20y$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

315

QUESTION ID

S9-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5y(y − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 2

Recognize perfect squares and perfect cubes • Grade 8 • Expressions & Equations • 8.EE.A.2

Write $16x^4$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

317

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(4x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

Write $8x^3$ as a cube.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

318

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x)^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

Which is a perfect cube?

A. 32

B. 16

C. 36

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

319

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 64

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

True or false?

x^6 is both a perfect square and a perfect cube.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

320

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x^3)^2$ and $(x^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the cube root of 27.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

321

QUESTION ID

S9-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 3

Multiply to check a factoring pattern • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(a + b)(a^2 - ab + b^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

323

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$a^3 + b^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 2)(x^2 + 2x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

324

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

True or false?

$(x + 3)(x^2 - 6x + 9) = x^3 + 27.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

325

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle term should be $-3x$: $(x + 3)(x^2 - 3x + 9) = x^3 + 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 9)(x^2 + 9)$ equals

A. $x^4 - 18x^2 - 81$

B. $x^4 + 81$

C. $(x - 3)^4$

D. $x^4 - 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

326

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^4 - 81$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

Multiply $(2x + 1)(4x^2 - 2x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

327

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$8x^3 + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 1

Factor expressions whose parts are themselves expressions • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $(x + 1)^2 - 9$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

329

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 4)$

GRADING NOTE Form required: factored completely. $((x + 1) - 3)((x + 1) + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(2x - 1)^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

330

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(4x^2 - 8x + 7)$

GRADING NOTE Form required: factored completely. a = 2x − 1, b = 2: $(a + b)(a^2 - 2a + 4) = (2x + 1)((2x - 1)^2 - 2(2x - 1) + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

True or false?

$a^6 - b^6$ can be factored first as a difference of squares or first as a difference of cubes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

331

QUESTION ID

S9-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^3)^2 - (b^3)^2$ or $(a^2)^3 - (b^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

$(x - y)^2 - (x + y)^2$ equals

A. 0

B. $4xy$

C. $-4xy$

D. $-2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

332

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-4xy$

GRADING NOTE $((x - y) - (x + y))((x - y) + (x + y)) = (-2y)(2x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

Factor $x^2(x + 2) - 4(x + 2)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

333

QUESTION ID

S9-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 2)^2(x - 2)$

GRADING NOTE Form required: factored completely. $(x + 2)(x^2 - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 2

Complete the factorization over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^4 - 81$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

335

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 3)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $16x^4 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

336

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(2x - 1)(2x + i)(2x - i)$

GRADING NOTE Any equivalent factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

True or false?

Over the complex numbers, $x^3 + 8$ has only one linear factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

337

QUESTION ID

S9-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has three: $(x + 2)$ and two from $x^2 - 2x + 4 = (x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

Which polynomial is fully factored over the integers but not over the complex numbers?

A. $x^2 + 9$

B. $x - 9$

C. $x^2 - 9$

D. $(x - 3)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

338

QUESTION ID

S9-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 9$

GRADING NOTE $x^2 + 9 = (x + 3i)(x - 3i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

$2x^5 - 32x = 2x(x^2 + 4)(x + 2)(x - 2)$ over the integers. Complete the factorization over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

339

QUESTION ID

S9-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x + 2i)(x - 2i)(x + 2)(x - 2)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2 • SEEING STRUCTURE IN EXPRESSIONS

Rewrite expressions in quadratic form and factor

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Factor quadratic trinomials	**HSA-SSE.A.2**
BACKWARD 2	Write powers as squares of other powers	**8.EE.A.1**
BACKWARD 3	Treat a repeated part as a single variable	**HSA-SSE.A.1.b**
FORWARD 1	Solve equations in quadratic form over the complex numbers	**HSN-CN.C.7**
FORWARD 2	Solve exponential equations in quadratic form	**HSF-LE.A.4**

SET 10 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 5x^2 + 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

342

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

Factor $x^4 + 3x^2 - 10$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

343

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 5)(x^2 - 2)$

GRADING NOTE Form required: factored completely over the integers ($x^2 - 2$ does not factor over the integers). $u^2 + 3u - 10 = (u + 5)(u - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

Factor $9^x - 4 \cdot 3^x + 3$ by writing it in quadratic form.
Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

344

QUESTION ID
S10-M-Q3

TYPE
Short answer

CORRECT ANSWER
$(3^x - 1)(3^x - 3)$

GRADING NOTE Form required: a product of two factors. $9^x = (3^x)^2$, so with $u = 3^x$: $u^2 - 4u + 3 = (u - 1)(u - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

Factor $x^6 - 9x^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

345

QUESTION ID

S10-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x^2 + x + 1)(x - 2)(x^2 + 2x + 4)$

GRADING NOTE Form required: factored completely. $u = x^3$: $(u - 1)(u - 8)$; then difference of cubes twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

Factor $2x^4 - 7x^2 - 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

346

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(2x^2 + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. $2u^2 - 7u - 4 = (2u + 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 1

Factor quadratic trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 5u + 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

348

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

Factor $u^2 + 3u - 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

349

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(u + 5)(u − 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

$2u^2 - 7u - 4$ equals

A. $(2u + 1)(u - 4)$

B. $(u - 2)(2u + 2)$

C. $(2u + 4)(u - 1)$

D. $(2u - 1)(u + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

350

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2u + 1)(u − 4)

GRADING NOTE Check: $2u^2 - 8u + u - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$u^2 - 9u + 8 = (u + 1)(u - 8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

351

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(u + 1)(u - 8) = u^2 - 7u - 8$; the correct factorization is $(u - 1)(u - 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 4u + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

352

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 2

Write powers as squares of other powers • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write x^4 as a power of x^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

354

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 9^x as a power of 3^x. Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

355

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^x)^2$

GRADING NOTE Any equivalent form. $9^x = (3^2)^x = (3^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

x^6 equals

A. $(x^4)^2$

B. $(x^3)^2$

C. $x^3 \cdot x^3 \cdot x^3$

D. $(x^2)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

356

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^3)^2$

GRADING NOTE $3 \cdot 2 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$4^x = (2^x)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

357

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = 2^{2x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 25^x as a power of 5^x. Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

358

QUESTION ID

S10-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5^x)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 3

Treat a repeated part as a single variable • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

In $x^4 - 5x^2 + 4$, what single variable u turns the expression into $u^2 - 5u + 4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

360

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$u = x^2$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2(x - 1)^2 + 3(x - 1) - 5$ using $u = x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

361

QUESTION ID

S10-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$2u^2 + 3u - 5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

True or false?

In $4^x - 2^x - 6$, the substitution $u = 2^x$ gives $u^2 - u - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

362

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = (2^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

Which substitution writes $x^6 - 9x^3 + 8$ as a quadratic?

A. $u = x^2$

B. $u = x^3$

C. $u = x^6$

D. $u = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

363

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $u = x^3$

GRADING NOTE $x^6 = (x^3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

After factoring $(u - 1)(u - 8)$ with $u = x^3$, write the result in terms of x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

364

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x^3 - 1)(x^3 - 8)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 1

Solve equations in quadratic form over the complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^4 + 3x^2 - 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

366

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1, -1, 2i, -2i$

GRADING NOTE Form required: all four. $x^2 = 1$ or $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

Solve $x^4 - 5x^2 + 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

367

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1 or −1 or 2 or −2

GRADING NOTE Form required: all four solutions, in any order. $(x^2 - 1)(x^2 - 4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

True or false?

$x^4 + 10x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

368

QUESTION ID

S10-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x^2 + 1)(x^2 + 9) = 0$ has only the solutions $\pm i$ and $\pm 3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

How many complex solutions, with multiplicity, does $x^4 - 13x^2 + 36 = 0$ have?

A. 1

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

369

QUESTION ID

S10-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE ±2 and ±3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

Solve $x^4 + x^2 - 12 = 0$ over the complex numbers. Give exact solutions.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

370

QUESTION ID

S10-F1-Q5

TYPE

Short answer

CORRECT ANSWER

√3 or −√3 or 2i or −2i

GRADING NOTE Form required: all four, exact, in any order. $(x^2 + 4)(x^2 - 3) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 2

Solve exponential equations in quadratic form • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $9^x - 5 \cdot 3^x + 6 = 0$. Give both solutions as decimals rounded to 3 decimal places.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

372

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.631 or 1

GRADING NOTE Form required: both, rounded as stated. $3^x = 2$ or $3^x = 3$, so $x = \log_3 2 = \ln 2 / \ln 3 \approx 0.631$ or $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

Solve $4^x - 6 \cdot 2^x + 8 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

373

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 1 or x = 2

GRADING NOTE Form required: both. $2^x = 2$ or $2^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

True or false?

$4^x + 2^x - 2 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

374

QUESTION ID

S10-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^x = 1$ gives $x = 0$; $2^x = -2$ is impossible since $2^x > 0$. One solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $25^x - 26 \cdot 5^x + 25 = 0$ are

A. 0 and 2

B. 1 and 25

C. 0 and 1

D. 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

375

QUESTION ID
S10-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 0 and 2

GRADING NOTE $5^x = 1$ or $5^x = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

Solve $e^{2t} - 7e^{t} + 10 = 0$. Give both solutions as decimals rounded to 3 decimal places.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

376

QUESTION ID

S10-F2-Q5

TYPE

Short answer

CORRECT ANSWER

0.693 or 1.609

GRADING NOTE Form required: both, rounded as stated. $e^{t} = 2$ or $e^{t} = 5$, so $t = \ln 2 \approx 0.693$ or $t = \ln 5 \approx 1.609$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4 • SEEING STRUCTURE IN EXPRESSIONS

Derive the formula for the sum of a finite geometric series

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Write terms of a geometric sequence	**HSF-LE.A.2**
BACKWARD 2	Distribute a factor over a sum	**6.EE.A.3**
BACKWARD 3	Use the product rule for exponents	**8.EE.A.1**
FORWARD 1	Prove the sum formula from a polynomial identity	**HSA-APR.C.4**
FORWARD 2	Describe how partial sums behave when the ratio is between -1 a...	**HSA-SSE.B.4**

SET 11 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

$S_n = a + ar + ar^2 + \ldots + ar^{n-1}$ with $r \neq 1$. In the derivation of $S_n = a(1 - r^n)/(1 - r)$, which equation results from subtracting rS_n from S_n?

A. $S_n - rS_n = a - ar^n$

B. $S_n - rS_n = ar - ar^n$

C. $S_n - rS_n = a - ar$

D. $S_n - rS_n = a - ar^{n-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

379

QUESTION ID

S11-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $S_n - rS_n = a - ar^n$

GRADING NOTE $rS_n = ar + ar^2 + \ldots + ar^n$, so every middle term cancels and $S_n - rS_n = a - ar^n$. Then $(1 - r)S_n = a(1 - r^n)$, and dividing by $1 - r$ (allowed because $r \neq 1$) gives the formula. The others use the wrong last term of rS_n, cancel too much, or lose the first term a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

$S = 3 + 6 + 12 + \ldots + 3 \cdot 2^9$. With the multiply-and-subtract method, $2S - S$ equals which expression, and what is S?

A. $3 \cdot 2^9 - 3$, so S = 1533

B. $3 \cdot 2^{10} + 3$, so S = 3075

C. $3 \cdot 2^{10} - 6$, so S = 3066

D. $3 \cdot 2^{10} - 3$, so S = 3069

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

380

QUESTION ID

S11-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $3 \cdot 2^{10} - 3$, so S = 3069

GRADING NOTE $2S = 6 + 12 + \ldots + 3 \cdot 2^{10}$; subtracting S leaves the last term of 2S minus the first term of S: $3 \cdot 2^{10} - 3 = 3072 - 3 = 3069$. The others use the wrong last term, subtract the wrong first term, or add it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

Why does the derivation of $S_n = a(1 - r^n)/(1 - r)$ fail when $r = 1$, and what is S_n then?

A. When $r = 1$ the terms a, ar, ar^2, ... grow without bound, so S_n does not exist and no formula for it is possible.

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

C. rS_n cannot be formed when $r = 1$, so $S_n = a$.

D. The formula gives 0/0, which means $S_n = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

381

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

GRADING NOTE Division by $1 - r = 0$ is not allowed; with $r = 1$ the series is $a + a + \ldots + a$ (n terms) $= na$. The others treat 0/0 as 0, misread the method, or confuse $r = 1$ with $|r| > 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

$T = 1 + x + x^2 + \ldots + x^n$ ($n + 1$ terms, $x \neq 1$).
Multiplying by x and subtracting T gives which formula?
(Check: $x = 2$, $n = 3$ gives $1 + 2 + 4 + 8 = 15$.)

A. $T = (x^n - 1)/(x - 1)$

B. $T = (x^{n+1} - x)/(x - 1)$

C. $T = (x^{n+1} - 1)/(x - 1)$

D. $T = (x^{n+1} - 1)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

382

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C.

$T = (x^{n+1} - 1)/(x - 1)$

GRADING NOTE $xT = x + x^2 + \ldots + x^{n+1}$, so $xT - T = x^{n+1} - 1$ and $T = (x^{n+1} - 1)/(x - 1)$. Check: $(16 - 1)/1 = 15$. The others give 7, 5, and 14 for the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

$T = 100 + 100(1.05) + 100(1.05)^2 + \ldots + 100(1.05)^{19}$ (20 yearly amounts). Using the multiply-and-subtract method with $1.05T - T$, which equation and total are correct? A calculator may be used.

A. $0.05T = 100(1.05^{19} - 1)$, so $T \approx \$3{,}053.90$

B. $1.05T = 100(1.05^{20} - 1)$, so $T \approx \$157.46$

C. $0.05T = 100(1.05^{21} - 1)$, so $T \approx \$3{,}571.93$

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

383

QUESTION ID

S11-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

GRADING NOTE $1.05T = 100(1.05) + \ldots + 100(1.05)^{20}$, so $1.05T - T = 100(1.05)^{20} - 100$ and $T = 100(1.05^{20} - 1)/0.05 \approx 3306.60$. The others use the wrong last power (19 or 21) or forget that $1.05T - T = 0.05T$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 1

Write terms of a geometric sequence • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the first four terms of the geometric sequence with first term 3 and ratio 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

385

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3, 6, 12, 24

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

What is the 10th term of 3, 6, 12, ...?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

386

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1536

GRADING NOTE $3 \cdot 2^9 = 1536$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

The nth term of a geometric sequence with first term a and ratio r is

A. $a + (n - 1)r$

B. $a \cdot n \cdot r$

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

387

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ar^{n-1}

GRADING NOTE The first term is ar^0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

True or false?

100, 105, 110.25, ... is arithmetic with common difference 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

388

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 5 and 5.25; the ratio 1.05 is constant, so it is geometric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the ratio of 81, 27, 9, 3, ...

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

389

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 2

Distribute a factor over a sum • Grade 6 • Expressions & Equations • 6.EE.A.3

Distribute: 2(3 + 6 + 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

391

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 + 12 + 24

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distribute: r(a + ar + ar²). Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

392

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$a \cdot r + a \cdot r^2 + a \cdot r^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

True or false?

$x(1 + x + x^2) = x + x^2 + x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

393

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

1.05(100 + 105) equals

A. 105 + 105

B. 100 + 110.25

C. 105 + 110.25

D. 205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

394

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 105 + 110.25

GRADING NOTE Each term is multiplied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Subtract: (6 + 12 + 24 + 48) − (3 + 6 + 12 + 24).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

395

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45

GRADING NOTE The shared terms cancel: 48 − 3 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 3

Use the product rule for exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $r \cdot r^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

397

QUESTION ID

S11-B3-Q1

TYPE

Short answer

CORRECT ANSWER

r^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $ar^{n-1} \cdot r$.

A. a^2r^n

B. ar^{n-2}

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

398

QUESTION ID

S11-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. ar^n

GRADING NOTE $r^{n-1} \cdot r = r^{n-1+1} = r^n$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2 \cdot 2^9$ equals

A. 2^{10}

B. 4^{10}

C. 4^9

D. 2^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

399

QUESTION ID

S11-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2^{10}

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

True or false?

$x \cdot x^n = x^{2n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

400

QUESTION ID

S11-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Add exponents: $x^1 \cdot x^n = x^{n+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

Evaluate 2^{10}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

401

QUESTION ID

S11-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1024

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 1

Prove the sum formula from a polynomial identity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity
$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$?

A. Divide both sides by $x - 1$ to get $(x^n - 1)/(x - 1) = x^{n-1} + \ldots + x + 1$; this is the geometric sum formula, so the identity is true.

B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

C. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and 1, leaving $x^n + 1$.

D. For $x = 2$ and $n = 3$: $7 = 1 \cdot 7$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

403

QUESTION ID
S11-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

GRADING NOTE Distribute and cancel (telescoping). The others check one example, assume what is to be proved (and divide by $x - 1$, which can be 0), or make a sign error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

Use the identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + 1)$ with $x = 3$ to write $1 + 3 + 9 + \ldots + 3^5$ as $(3^m - 1)/(3 - 1)$. Give m and the value of the sum.

m = _____

Sum = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

404

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6; 364

GRADING NOTE Form required: both. $(3^6 - 1)/2 = 728/2 = 364$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

True or false?

The identity shows that $x - 1$ is a factor of $x^n - 1$ for every whole number $n \geq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

405

QUESTION ID

S11-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It writes $x^n - 1$ as $(x - 1)$ times a polynomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

Which equals $1 + 2 + 4 + \ldots + 2^9$?

A. 2^{10}

B. $2^{10} - 1$

C. $2^9 - 1$

D. $(2^9 - 1)/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

406

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2^{10} - 1$

GRADING NOTE $(2^{10} - 1)/(2 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

The identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + x + 1)$ holds for every x. How does it give $a + ar + \ldots + ar^{n-1} = a(1 - r^n)/(1 - r)$ for $r \neq 1$?

A. Put x = a and divide both sides by r: the sum is $(a^n - 1)/(a - 1)$, which is the same as $a(1 - r^n)/(1 - r)$ after simplifying.

B. Put x = 1: both sides of the identity are 0, so the identity is checked at that value, and therefore the sum formula holds for every ratio $r \neq 1$.

C. Put x = r and add a to both sides of the identity: the sum is $r^n - 1 + a$, which equals $a(1 - r^n)/(1 - r)$ for $r \neq 1$.

D. Put x = r and multiply by a: $a(r^n - 1) = a(r - 1)(1 + r + \ldots + r^{n-1})$; divide by $r - 1$ ($r \neq 1$), then multiply top and bottom by -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

407

QUESTION ID

S11-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Put x = r and multiply by a: $a(r^n - 1) = a(r - 1)(1 + r + \ldots + r^{n-1})$; divide by $r - 1$ ($r \neq 1$), then multiply top and bottom by -1.

GRADING NOTE Substitute x = r, scale by a, divide by r − 1. The others substitute the wrong variable, use an invalid operation, or use the value x = 1, which the formula excludes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 2

Describe how partial sums behave when the ratio is between −1 and 1 • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

For the series 8 + 4 + 2 + 1 + ..., find S_5 and S_{10} (the sums of the first 5 and first 10 terms), to 4 decimal places.

S_5 = _____

S_{10} = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

409

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15.5000; 15.9844

GRADING NOTE Rounding to 4 decimal places. $S_n = 8(1 - 0.5^n)/0.5 = 16(1 - 0.5^n)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

$S_n = 16(1 - 0.5^n)$ is the sum of the first n terms of 8 + 4 + 2 + ... Why is every partial sum less than 16, yet as close to 16 as you like for large n?

A. 0.5^n reaches 0 when n = 16, so the partial sums are less than 16 until the 16th term and equal 16 after it.

B. 0.5^n is always positive, so $1 - 0.5^n < 1$ and $S_n < 16$; 0.5^n shrinks toward 0 as n grows, so S_n gets as close to 16 as you like.

C. 0.5^n grows as n grows, so S_n approaches 16 from above.

D. $S_{10} \approx 15.98$ is less than 16 and very close to it, so every partial sum is less than 16 and as close as you like.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

410

QUESTION ID

S11-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0.5^n is always positive, so $1 - 0.5^n < 1$ and $S_n < 16$; 0.5^n shrinks toward 0 as n grows, so S_n gets as close to 16 as you like.

GRADING NOTE Treat 0.5^n as one positive quantity that shrinks toward 0. The others claim it grows or reaches 0, or reason from one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

True or false?

For r = 2, the partial sums of $a + ar + ar^2 + \ldots$ ($a > 0$) stay below a fixed number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

411

QUESTION ID

S11-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE r^n grows without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

For 10 − 5 + 2.5 − 1.25 + …, the partial sums get close to

A. 20/3

B. 15

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

412

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20/3

GRADING NOTE a/(1 − r) = 10/1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

A ball falls 10 m, then rebounds to 60% of each previous height, so the total distance after n bounces is $10 + 2 \cdot 6(1 - 0.6^n)/0.4$ meters. What total distance (in m) can it approach but never reach?

Distance (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

413

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

40

GRADING NOTE Form required: the value. $2 \cdot 6/0.4 = 30$ and $10 + 30 = 40$; since $0.6^n > 0$, the factor $1 - 0.6^n$ is always less than 1, so the total stays below 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

Use the finite geometric series formula to solve problems

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Identify the first term, ratio, and number of terms	**HSF-LE.A.2**
BACKWARD 2	Change a percent to a growth factor	**7.RP.A.3**
BACKWARD 3	Round results of calculations correctly	**HSN-Q.A.3**
FORWARD 1	Find the number of terms with logarithms	**HSF-LE.A.4**
FORWARD 2	Compare two payment plans with the series formula	**HSA-SSE.B.4**

SET 12 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 12 terms of 5 + 15 + 45 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

416

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

1,328,600

GRADING NOTE Any equivalent form. $5(3^{12} - 1)/(3 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

Leah deposits $250 at the end of each month into an account that earns 0.5% interest per month. The balance just after the 36th deposit is $250 + 250(1.005) + \ldots + 250(1.005)^{35}$. Find it to the nearest cent.

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

417

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

$9834.03

GRADING NOTE Rounding to the nearest cent required. $250(1.005^{36} - 1)/0.005 = 9834.03$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

A $180,000 loan is repaid with 360 equal monthly payments P at a monthly interest rate of 0.4%. P satisfies
$180{,}000 = P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$. Use the geometric series formula to find P to the nearest cent.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

QUESTION ID
S12-M-Q3

TYPE
Short answer

CORRECT ANSWER
$944.40

GRADING NOTE Rounding to the nearest cent required. a = P/1.004, r = 1/1.004, n = 360: P = 180,000 · $0.004/(1 - 1.004^{-360}) = 944.40$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

A ball is dropped from 10 m and each rebound reaches 60% of the previous height. Find the total vertical distance it has traveled when it hits the ground for the 6th time, to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

419

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

37.67 m

GRADING NOTE Rounding to 0.01 required. $10 + 2(6 + 3.6 + 2.16 + 1.296 + 0.7776) = 10 + 2 \cdot 6(1 - 0.6^5)/0.4 = 37.6672$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

The sum of the first n terms of 2 + 6 + 18 + ... is $S_n = 3^n - 1$. What is the least number of terms whose sum is greater than 10,000? Give that number and the sums of the first 8 and the first 9 terms.

Least number of terms: _____

Sum of the first 8 terms: _____

Sum of the first 9 terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

420

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

9; 6,560; 19,682

GRADING NOTE Form required: all three. $S_8 = 3^8 - 1 = 6{,}560 \le 10{,}000$ and $S_9 = 3^9 - 1 = 19{,}682 > 10{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 1

Identify the first term, ratio, and number of terms • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

For the geometric series 5 + 15 + 45 + ... with 12 terms, give the first term a, the ratio r, and the number of terms n.

a = _____

r = _____

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

422

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5; 3; 12

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B1-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $250 + 250(1.005) + \ldots + 250(1.005)^{35}$, give the first term a, the ratio r, and the number of terms n.

a = _____

r = _____

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

423

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

250; 1.005; 36

GRADING NOTE Form required: all three. The exponents run from 0 to 35, which is 36 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B1-Q2**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

For 6 + 3.6 + 2.16 + 1.296 + 0.7776, r is

A. 0.4

B. 1.67

C. 0.6

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

424

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.6

GRADING NOTE 3.6/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A list with exponents 0 through 35 has 35 terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

425

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 36 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

For $P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$, which gives the first term a and the common ratio r?

A. a = P/1.004, r = 1/1.004

B. a = 1/1.004, r = P

C. a = P, r = 1.004

D. a = P/1.004, r = 1.004

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

426

QUESTION ID
S12-B1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. a = P/1.004, r = 1/1.004

GRADING NOTE Each term is the previous one divided by 1.004.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 2

Change a percent to a growth factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What factor adds 0.5% interest?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

428

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.005

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A height is 60% of the one before. What is the ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

429

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

0.4% per month as a factor is

A. 1.04

B. 1.004

C. 0.004

D. 1.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

430

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.004

GRADING NOTE 1 + 0.004.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

True or false?

Interest of 12% per year is a factor of 0.12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

431

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The growth factor is 1 + 0.12 = 1.12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A loss of 15% per year is what factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.85

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 3

Round results of calculations correctly • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 9834.0262 to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

434

QUESTION ID

S12-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9834.03

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 37.6672 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

435

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

37.67

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

True or false?

Rounding 1.005 to 1.01 before raising it to the 36th power gives the same answer to the nearest cent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

436

QUESTION ID

S12-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Early rounding changes the result a lot; round only at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

1.005^{36} is closest to

A. 1.1967

B. 1.18

C. 1.2

D. 1.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

437

QUESTION ID

S12-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1.1967

GRADING NOTE Calculator value 1.19668....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 3^{12} exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

438

QUESTION ID

S12-B3-Q5

TYPE

Short answer

CORRECT ANSWER

531441

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 1

Find the number of terms with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

How many end-of-month deposits of $250 at 0.5% per month are needed for the balance to reach $20,000? Use $250(1.005^n - 1)/0.005 \geq 20{,}000$ and give the smallest whole number n.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

68

GRADING NOTE $1.005^n \geq 1 + 20{,}000(0.005)/250 = 1.4$, so $n \geq \ln 1.4/\ln 1.005 \approx 67.46$; check: n = 67 gives about $19,838.70 and n = 68 about $20,187.89.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

The sum of the first n terms of $2 + 6 + 18 + \ldots$ is $3^n - 1$. Solve $3^n - 1 = 1{,}000{,}000$ for n (to 2 decimal places), then give the least whole number of terms whose sum is at least 1,000,000.

n (2 decimal places) = _____

Least number of terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

441

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12.58; 13

GRADING NOTE Form required: both. $n = \log_3(1{,}000{,}001) = \ln(1{,}000{,}001)/\ln 3 \approx 12.575$; 12 terms give 531,440, 13 terms give 1,594,322.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

True or false?

For a ratio greater than 1, more terms always give a larger sum when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

442

QUESTION ID
S12-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each added term is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

The sum $5(2^n - 1)$ first exceeds 5,000 when n =

A. 9

B. 11

C. 10

D. 1000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

443

QUESTION ID

S12-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE $2^n > 1001$; $2^{10} = 1024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

The sum of a geometric series with positive terms first reaches a target when n ≈ 12.58. Why must n be rounded up to 13 rather than simply rounded to the nearest whole number?

A. Rounding down to 12 is safer: fewer terms always give a sum that still reaches the target.

B. Answers must always be rounded to the nearest whole number, and 12.58 rounds to 13 that way.

C. Any whole number near 12.58 gives about the same sum, so 12 and 13 are both acceptable answers, and the choice is a matter of style.

D. The sum increases with n and must reach the target; 12 terms give a sum below it, so round up to 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

444

QUESTION ID

S12-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The sum increases with n and must reach the target; 12 terms give a sum below it, so round up to 13.

GRADING NOTE Only n ≥ 12.58 meets the target, so the least whole number is 13 (rounding up). The others rely on ordinary rounding (12.4 would round to 12, which fails), prefer a count that fails, or accept both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 2

Compare two payment plans with the series formula • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Plan A pays $1,000 per month for 24 months. Plan B pays $500 the first month and 8% more each month after that, for 24 months. Find Plan B's total and how much more than Plan A it pays, to the nearest cent.

Plan B total ($): _____

Difference ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

446

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

33,382.38; 9,382.38

GRADING NOTE Rounding to the nearest cent. Plan B: $500(1.08^{24} - 1)/0.08 \approx 33{,}382.38$; Plan A: 24,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

Plan A pays $1,000 per month. Plan B pays $500 the first month and 8% more each month after that. After how many months does Plan B's total first exceed Plan A's total for the same number of months?

Number of months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

447

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Form required: the whole number. Compare $500(1.08^n - 1)/0.08$ with $1000n$: $n = 17$ gives $16{,}875.11 < 17{,}000$; $n = 18$ gives $18{,}725.12 > 18{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

True or false?

A geometric plan with ratio greater than 1 eventually pays more in total than any plan with a fixed monthly amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

448

QUESTION ID
S12-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its terms grow without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

Plan C pays $2,000 then 5% less each month. Its total over 24 months is closest to

A. $40,000

B. $24,000

C. $48,000

D. $28,320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

449

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $28,320

GRADING NOTE $2000(1 - 0.95^{24})/0.05 \approx 28{,}320$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

Plan A pays $1,000 per month for 24 months; Plan B pays $500 the first month and 8% more each month after that. Comparing the plans only by their 24-month totals assumes which of the following?

A. That both plans pay exactly $24,000 in total over the 24 months.

B. That Plan B pays the same amount every month, just like Plan A.

C. That money received later is worth the same as money now (no interest or inflation).

D. That the 8% increase in Plan B is applied only once, in month 2, and not again after that.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

450

QUESTION ID

S12-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. That money received later is worth the same as money now (no interest or inflation).

GRADING NOTE Totals ignore when the money arrives. The others contradict the plans as stated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

Add and subtract polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Combine like terms with the same power	**7.EE.A.1**
BACKWARD 2	Subtract a whole expression	**6.EE.A.3**
BACKWARD 3	Write polynomials in standard form	**HSA-SSE.A.1.a**
FORWARD 1	Reason about the degree of a sum or difference	**HSA-APR.A.1**
FORWARD 2	Add and subtract rational expressions	**HSA-APR.D.7**

SET 13 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Simplify $(4x^3 - 2x^2 + 7) + (-x^3 + 5x - 9)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

453

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3 - 2x^2 + 5x - 2$

GRADING NOTE Form required: standard form (descending powers). $4x^3 - x^3 = 3x^3$; $-2x^2$; $5x$; $7 - 9 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

Simplify $(6x^4 - x^3 + 2x) - (2x^4 + 3x^3 - x^2 + 2x - 1)$.
Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

454

QUESTION ID
S13-M-Q2

TYPE
Short answer

CORRECT ANSWER
$4x^4 - 4x^3 + x^2 + 1$

GRADING NOTE Form required: standard form. The x terms cancel: $2x - 2x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

A company's revenue is $R(x) = -0.01x^3 + 3x^2 + 40x$ dollars and its cost is $C(x) = 0.02x^2 + 15x + 900$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ in standard form.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

455

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-0.01x^3 + 2.98x^2 + 25x - 900$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

Subtract $(x^5 + 2x^4 - 3x^3 - 7)$ from $(x^5 - 3x^3 + x)$. Combine like terms and write the result in standard form; then give its degree.

Result: _____

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

456

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

$-2x^4 + x + 7$; 4

GRADING NOTE Form required: both. $(x^5 - 3x^3 + x) - (x^5 + 2x^4 - 3x^3 - 7)$: the x^5 and x^3 terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

$p(x) = 5x^3 + x^2 - 6$. Find the polynomial $q(x)$ for which $p(x) + q(x) = 2x^3 - x + 4$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

457

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-3x^3 - x^2 - x + 10$

GRADING NOTE Form required: standard form. $q = (2x^3 - x + 4) - (5x^3 + x^2 - 6)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 1

Combine like terms with the same power • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $4x^3 - x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

459

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-2x^2 + 5x - 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

460

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 + 5x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

True or false?

$x^3 + x^2 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

461

QUESTION ID

S13-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^3 and x^2 are unlike terms; they cannot be combined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

Which pair are like terms?

A. $4x^2$ and $-x^2$

B. $4x^2$ and $4x$

C. x^3 and $3x$

D. x and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

462

QUESTION ID

S13-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2$ and $-x^2$

GRADING NOTE Same variable, same power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $7 - 9 + 2x - 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

463

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 2

Subtract a whole expression • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $-(2x^4 + 3x^3 - x^2 + 2x - 1)$ without parentheses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

465

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^4 - 3x^3 + x^2 - 2x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (5 − x) − (3 − 4x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

466

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3x + 2

GRADING NOTE Any equivalent simplified form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

True or false?

$a - (b - c) = a - b - c.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

467

QUESTION ID

S13-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $a - (b - c) = a - b + c.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

$-(x^2 - 7)$ equals

A. $x^2 + 7$

B. $-x^2 - 7$

C. $-x^2 + 7$

D. $x^2 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

468

QUESTION ID

S13-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^2 + 7$

GRADING NOTE Distribute -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 10 − (x + 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

469

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6 − x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 3

Write polynomials in standard form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write $7 + 5x - 2x^3 + x^2$ in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

471

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^3 + x^2 + 5x + 7$

GRADING NOTE Form required: descending powers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

What is the leading coefficient of $3 - 4x^2 + x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

472

QUESTION ID

S13-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

The degree of $-2x^4 + x + 7$ is

A. 7

B. 1

C. 4

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

473

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE Highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

True or false?

The standard form of $2x - x^3$ is $2x - x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

474

QUESTION ID

S13-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Standard form lists powers in descending order: $-x^3 + 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

What is the constant term of $6x^3 - 2x + 9$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

475

QUESTION ID

S13-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 1

Reason about the degree of a sum or difference • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

p(x) and q(x) both have degree 4. Which statement about p(x) + q(x) is true?

A. Its degree is 4 or less, or it is the zero polynomial: $(x^4 + x) + (-x^4 + 1) = x + 1$.

B. Its degree is always exactly 4, because adding never changes the highest power.

C. Its degree is always 4 or more, because adding can only raise the degree.

D. Its degree is always 8, because adding polynomials adds their degrees: 4 + 4 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

477

QUESTION ID
S13-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Its degree is 4 or less, or it is the zero polynomial: $(x^4 + x) + (-x^4 + 1) = x + 1$.

GRADING NOTE Adding cannot create higher powers, but the x^4 terms can cancel. The others add the degrees (that happens for products), or ignore cancellation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

For which value of k does
$(kx^3 + 2x - 1) - (5x^3 - x^2 + 3)$ have degree 2?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

478

QUESTION ID
S13-F1-Q2

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form. The x^3 terms cancel when k = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

479

QUESTION ID
S13-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Polynomials are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

p has degree 5 and q has degree 3. The degree of $p - q$ is

A. 5

B. 2

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

480

QUESTION ID
S13-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 5

GRADING NOTE The x^5 term of p cannot cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

Why is the sum of two polynomials always a polynomial?

A. The sum always has a higher degree than either polynomial, and anything with a higher degree is a polynomial.

B. Adding combines like terms, and each term is still a real number times a whole-number power of x.

C. Adding powers of x gives a power of x, as $x^2 + x^3 = x^5$ shows, so each term of the sum is still a power of x.

D. Any expression that contains x, such as 1/x or √x, is a polynomial, so every sum of such expressions is a polynomial too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

481

QUESTION ID
S13-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Adding combines like terms, and each term is still a real number times a whole-number power of x.

GRADING NOTE Closure under addition: like terms combine and no negative or fractional exponents appear. The others give false statements (degrees do not increase; 1/x is not a polynomial; $x^2 + x^3 \neq x^5$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 2

Add and subtract rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Write 3/x + 2/(x + 1) as one fraction with denominator x(x + 1). Give the numerator and the excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

483

QUESTION ID
S13-F2-Q1

TYPE
Short answer

CORRECT ANSWER
5x + 3; {0, −1}

GRADING NOTE Form required: both. 3(x + 1) + 2x = 5x + 3; x = 0 and x = −1 make a denominator zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

Write 1/(x − 2) − 1/(x + 2) as one fraction with denominator (x − 2)(x + 2). Give the numerator and the excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

484

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4; {2, −2}

GRADING NOTE Form required: both. (x + 2) − (x − 2) = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

True or false?

Polynomials are closed under division.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

485

QUESTION ID
S13-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For example 1 ÷ x is not a polynomial; rational expressions are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

x/2 + x/3 equals

A. 5x/5

B. x/5

C. 5x/6

D. 2x/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

486

QUESTION ID
S13-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 5x/6

GRADING NOTE Common denominator 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

How is adding rational expressions like adding fractions of integers?

A. In both, the numerators are multiplied together and the denominators are added together.

B. In both, the numerators are added and the denominators are added.

C. Rational expressions never need a common denominator, unlike fractions of integers, which always do.

D. Both are rewritten over a common denominator and then the numerators are added; polynomials play the role of integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

487

QUESTION ID
S13-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. Both are rewritten over a common denominator and then the numerators are added; polynomials play the role of integers.

GRADING NOTE Common denominator, then add numerators. The others use the error a/b + c/d = (a + c)/(b + d) or other invalid rules.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

Multiply polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Distribute a monomial over a polynomial	**HSA-APR.A.1**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
BACKWARD 3	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Find a coefficient of a product without full expansion	**HSA-APR.A.1**
FORWARD 2	Expand powers of binomials with the Binomial Theorem	**HSA-APR.C.5**

SET 14 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x^2 + 3x - 2)(2x - 5)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

490

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + x^2 - 19x + 10$

GRADING NOTE Form required: standard form. $2x^3 - 5x^2 + 6x^2 - 15x - 4x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

Expand $(x - 2)^3$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

491

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 6x^2 + 12x - 8$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

Multiply $(x^2 - x + 1)(x^2 + x + 1)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

492

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^4 + x^2 + 1$

GRADING NOTE Form required: standard form. $((x^2 + 1) - x)((x^2 + 1) + x) = (x^2 + 1)^2 - x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

Multiply (2x + 1)(x − 3)(x + 4). Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

493

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 3x^2 - 23x - 12$

GRADING NOTE Form required: standard form. $(x - 3)(x + 4) = x^2 + x - 12$; times $(2x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

A box has length x + 2, width x − 1, and height 2x + 3 (in cm). Write its volume as a polynomial in standard form.

V(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

494

QUESTION ID

S14-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 5x^2 - x - 6$

GRADING NOTE Form required: standard form. $(x^2 + x - 2)(2x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 1

Distribute a monomial over a polynomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $2x(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

496

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 6x^2 - 4x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

Multiply $-5(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

497

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 - 15x + 10$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

True or false?

$x^2(x + 4) = x^3 + 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

498

QUESTION ID

S14-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 \cdot 4 = 4x^2$, so the product is $x^3 + 4x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

$-3x(x - 1)$ equals

A. $3x^2 - 3x$

B. $-3x^2 + 3x$

C. $-3x^2 - 3x$

D. $-3x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

499

QUESTION ID

S14-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $-3x^2 + 3x$

GRADING NOTE $-3x \cdot -1 = 3x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $x(x^2 - x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

500

QUESTION ID

S14-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - x^2 + x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $x^2 \cdot x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

502

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $2x \cdot 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

503

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$x^2 \cdot x^2 = x^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

504

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$-x \cdot x^3$ equals

A. x^4

B. $-x^3$

C. $-x^4$

D. $-x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

505

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^4$

GRADING NOTE $-1 \cdot x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2)(-4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

506

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-4x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 3

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 3)(x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

508

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 12$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x + 2)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

509

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 2)^2$ equals

A. $x^2 + 4$

B. $x^2 - 4$

C. $x^2 - 4x + 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

510

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2 - 4x + 4$

GRADING NOTE Square of a binomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

True or false?

$(2x + 1)(x - 3) = 2x^2 - 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

511

QUESTION ID

S14-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle terms $-6x + x = -5x$ are missing: $2x^2 - 5x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

Multiply $(2x + 3)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

512

QUESTION ID

S14-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 + x - 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 1

Find a coefficient of a product without full expansion • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find the coefficient of x^2 in $(x^2 + 3x - 2)(4x^2 - x + 5)$ without expanding everything.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

514

QUESTION ID
S14-F1-Q1

TYPE
Short answer

CORRECT ANSWER
−6

GRADING NOTE Any equivalent form. x^2 terms: $x^2 \cdot 5 + 3x \cdot (-x) + (-2) \cdot 4x^2 = 5 - 3 - 8 = -6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

For which k does $(x + k)(x^2 - 3x + 1)$ have no x^2 term?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

515

QUESTION ID
S14-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. x^2 coefficient: $-3 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

True or false?

The constant term of a product of polynomials is the product of their constant terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

516

QUESTION ID
S14-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Only constants multiply to a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

The leading coefficient of $(2x^3 - 1)(5x^2 + x)$ is

A. 7

B. 5

C. 2

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

517

QUESTION ID
S14-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 10

GRADING NOTE $2 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

Find the coefficient of x in (x − 1)(x − 2)(x − 3).

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

518

QUESTION ID
S14-F1-Q5

TYPE
Short answer

CORRECT ANSWER
11

GRADING NOTE Any equivalent form. 1·2 + 1·3 + 2·3 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q5**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 1)^4$ using Pascal's Triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

520

QUESTION ID
S14-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^4 + 4x^3 + 6x^2 + 4x + 1$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

Expand $(x - 2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

521

QUESTION ID
S14-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x^4 - 8x^3 + 24x^2 - 32x + 16$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

The coefficients of $(x + y)^5$ are 1, 5, 10, 5, 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

522

QUESTION ID
S14-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Row 5 of Pascal's Triangle has six entries: 1, 5, 10, 10, 5, 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

The coefficient of x^2y^2 in $(x + y)^4$ is

A. 2

B. 6

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

523

QUESTION ID
S14-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 6

GRADING NOTE C(4, 2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

Why are the coefficients of $(x + y)^n$ symmetric (the same read forward and backward)?

A. C(n, k) = C(n, n − k): choosing which k of the n factors give x is the same as choosing which n − k give y.

B. x and y always have equal values in the expansion, so the coefficients of matching terms are the same.

C. Every coefficient of $(x + y)^n$ equals 1.

D. The middle coefficient is always the largest one, so the list of coefficients must be symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

524

QUESTION ID
S14-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. C(n, k) = C(n, n − k): choosing which k of the n factors give x is the same as choosing which n − k give y.

GRADING NOTE The count C(n, k) is symmetric. The others are false ($(x + y)^2 = x^2 + 2xy + y^2$) or do not explain the symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

Find remainders with the Remainder Theorem

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Evaluate a polynomial function	**HSF-IF.A.2**
BACKWARD 2	Powers of negative numbers	**7.NS.A.2**
BACKWARD 3	Dividend = divisor × quotient + remainder	**4.NBT.B.6**
FORWARD 1	Find unknown coefficients from given remainders	**HSA-APR.B.2**
FORWARD 2	Prove the Remainder Theorem	**HSA-APR.B.2**

SET 15 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Use the Remainder Theorem to find the remainder when $p(x) = x^3 - 4x^2 + 2x - 5$ is divided by $x - 3$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

527

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form. $p(3) = 27 - 36 + 6 - 5 = -8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

Use the Remainder Theorem to find the remainder when $p(x) = 2x^4 + x^3 - 3x + 7$ is divided by $x + 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

528

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

37

GRADING NOTE Any equivalent form. Divisor $x - (-2)$: $p(-2) = 32 - 8 + 6 + 7 = 37$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

Use the Remainder Theorem to find the remainder when $p(x) = x^{100} + 3$ is divided by $x + 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

529

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(-1) = (-1)^{100} + 3 = 1 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

Use the Remainder Theorem to find the remainder when $p(x) = 4x^3 - 6x^2 + x - 1$ is divided by $x - 1/2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

530

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form (−1.5). p(1/2) = 1/2 − 3/2 + 1/2 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 5x^2 + 4x$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

531

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(2) = 16 - 20 + 8 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 1

Evaluate a polynomial function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 - 4x^2 + 2x - 5$. Find $p(1)$.

$p(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

533

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

$p(x) = 2x^2 - x + 3$. Find $p(-2)$.

$p(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

534

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 8 + 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

True or false?

If $p(x) = x^2 + 1$, then $p(-3) = -8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

535

QUESTION ID

S15-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

If $p(x) = x^3 - x$, then $p(2) =$

A. 4

B. 6

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

536

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE 8 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

$p(x) = 4x - 1$. Find $p(1/2)$.

$p(1/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

537

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 2

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2

Evaluate $(-2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

539

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

Evaluate $(-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

540

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

$(-1)^{100}$ equals

A. -100

B. -1

C. 100

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

541

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE An even power of -1 is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

True or false?

$-2^4 = (-2)^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

542

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2^4 = -16$ but $(-2)^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2(-2)^4 + (-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

543

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. 32 − 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 3

Dividend = divisor × quotient + remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Write 47 = 5 × q + r with 0 ≤ r < 5.

q = _____

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

545

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9; 2

GRADING NOTE Form required: both. 47 = 5 × 9 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

A division gives quotient 12 and remainder 3 with divisor 7. What was the dividend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

546

QUESTION ID

S15-B3-Q2

TYPE

Short answer

CORRECT ANSWER

87

GRADING NOTE Any equivalent form. 7 × 12 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

If a division leaves remainder 0, the divisor is a factor of the dividend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

547

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dividend = divisor × quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

$x^3 - 1 = (x - 1)(x^2 + x + 1) + 0$. The remainder on division by $x - 1$ is

A. 0

B. 1

C. $x^2 + x + 1$

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

548

QUESTION ID

S15-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE Read from the identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

$p(x) = (x - 3)(x^2 - x - 1) - 8$. What is the remainder when $p(x)$ is divided by $x - 3$?

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

549

QUESTION ID

S15-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 1

Find unknown coefficients from given remainders • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

When $p(x) = x^3 + kx^2 - 4$ is divided by $x - 2$, the remainder is 12. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

551

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $p(2) = 8 + 4k - 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 + ax + b$ leaves remainder 5 on division by $x - 1$ and remainder -1 on division by $x + 1$. Find a and b.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

552

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2; 2

GRADING NOTE Form required: both. $p(1) = 1 + a + b = 5$ and $p(-1) = -1 - a + b = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

True or false?

If p(x) leaves remainder 0 on division by x − 4, then p(−4) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

553

QUESTION ID

S15-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The remainder on division by x − 4 is p(4), so p(4) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

$p(x) = 2x^2 + kx + 1$ leaves remainder 7 on division by $x + 1$. Then $k =$

A. 4

B. -4

C. -6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

554

QUESTION ID

S15-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE $2 - k + 1 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

For which k is the remainder of $x^4 - kx + 6$ divided by $x + 2$ equal to 0?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

555

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−11

GRADING NOTE Any equivalent form. 16 + 2k + 6 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 2

Prove the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Which argument proves that the remainder when a polynomial p(x) is divided by x − a is p(a)?

A. Write p(x) = (x − a)q(x) + r, where r is a constant because it has degree less than 1. Substituting x = a gives p(a) = 0 · q(a) + r = r.

B. Dividing x^3 by x − 2 leaves remainder 8, and $2^3 = 8$, so the remainder is always p(a).

C. Since x − a = 0 exactly when x = a, dividing p(x) by x − a is the same as replacing x with a, so the remainder must be p(a).

D. Write p(x) = (x − a)q(x) + r. Substituting x = 0 gives p(0) = (0 − a)q(0) + r = −a · q(0) + r, so the remainder r equals p(a) for every a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

QUESTION ID

S15-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Write p(x) = (x − a)q(x) + r, where r is a constant because it has degree less than 1. Substituting x = a gives p(a) = 0 · q(a) + r = r.

GRADING NOTE The division identity with a constant remainder, then x = a makes the factor (a − a) zero. The others check one example, substitute the wrong value (x = 0 gives p(0), not p(a)), or state the result without a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

When a polynomial p(x) is divided by x − a, why must the remainder be a constant?

A. Because dividing by x − a always comes out evenly, so the remainder always equals 0.

B. Because a is a constant, everything left over after dividing by x − a must also be a constant.

C. The remainder has a lower degree than the divisor x − a (degree 1), so it is a constant or 0.

D. Because p(x) always has degree 1 when it is divided by x − a, so the remainder is a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

558

QUESTION ID

S15-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The remainder has a lower degree than the divisor x − a (degree 1), so it is a constant or 0.

GRADING NOTE A remainder has smaller degree than the divisor. The others are false ($x^2 \div (x - 1)$ leaves 1) or give a non-reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

True or false?

The Remainder Theorem also works for division by 2x − 1 with remainder p(1/2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

559

QUESTION ID

S15-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE p(x) = (2x − 1)q(x) + r gives p(1/2) = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

Which step uses the fact that a − a = 0?

A. Graphing p(x) and reading its zeros

B. Dividing p(x) by x to find the quotient q(x)

C. Factoring p(x) into linear factors

D. Substituting x = a into (x − a)q(x) + r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

560

QUESTION ID

S15-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Substituting x = a into (x − a)q(x) + r

GRADING NOTE The product term vanishes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

$p(x) = (x - a)q(x) + r$, where r is a constant. Why is $p(a) = 0$ exactly when $x - a$ is a factor of $p(x)$?

A. $p(a) = 0$ for every polynomial written as $(x - a)q(x) + r$, so $x - a$ is always a factor.

B. Substituting $x = a$ gives $p(a) = r$; and $r = 0$ exactly when $p(x) = (x - a)q(x)$, which means $x - a$ is a factor.

C. $x - a$ is a factor exactly when $q(a) = 0$.

D. Substituting $x = a$ gives $p(a) = q(a)$, so $p(a) = 0$ exactly when $q(x)$ is a factor of $p(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

561

QUESTION ID
S15-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Substituting x = a gives p(a) = r; and r = 0 exactly when p(x) = (x − a)q(x), which means x − a is a factor.

GRADING NOTE The remainder equals p(a). The others are false statements about p(a) and q(a).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

Use p(a) = 0 to find factors

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
BACKWARD 2	Factor the quadratic quotient	**HSA-SSE.A.2**
BACKWARD 3	Divide a polynomial by a linear factor	**HSA-APR.A.1**
FORWARD 1	Choose parameters so that a given binomial is a factor	**HSA-APR.B.2**
FORWARD 2	Finish factorizations over the complex numbers	**HSN-CN.C.8**

SET 16 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

$p(x) = x^3 - 3x^2 - 4x + 12$ has $p(2) = 0$, so $x - 2$ is a factor. Factor $p(x)$ completely.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

564

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)(x - 3)$

GRADING NOTE Form required: the complete factorization. $p(2) = 8 - 12 - 8 + 12 = 0$; quotient $x^2 - x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

$p(x) = x^3 + 2x^2 - 5x - 6$ has $p(-1) = 0$, so $x + 1$ is a factor. Factor $p(x)$ completely.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

565

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1)(x + 3)(x - 2)$

GRADING NOTE Form required: the complete factorization. Quotient $x^2 + x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

$p(x) = 2x^3 + 5x^2 - 4x - 3$ has $p(-3) = 0$, so $x + 3$ is a factor. Factor $p(x)$ completely.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

566

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(2x + 1)(x - 1)$

GRADING NOTE Form required: the complete factorization. $p(-3) = -54 + 45 + 12 - 3 = 0$; quotient $2x^2 - x - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

Is $x - 4$ a factor of $p(x) = x^3 - 5x^2 + 2x + 9$? Use the Remainder Theorem: find $p(4)$, then answer Yes or No.

$p(4) =$ _____

Factor? (Yes or No): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

567

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

1; No

GRADING NOTE Form required: both. $p(4) = 64 - 80 + 8 + 9 = 1 \neq 0$, so $x - 4$ is not a factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

$p(x) = x^3 - 7x + 6$. Use $p(a) = 0$ to find all the zeros of p. List them separated by commas.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

568

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

{1, 2, −3}

GRADING NOTE Form required: all three zeros, any order. $p(1) = 0$, then the quotient $x^2 + x - 6 = (x - 2)(x + 3)$, so $p(x) = (x - 1)(x - 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 1

Use the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^3 - 3x^2 - 4x + 12$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

570

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

Find the remainder when $x^3 + 2x^2 - 5x - 6$ is divided by $x - 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

571

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

True or false?

Dividing by x + 3 means evaluating at x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

572

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + 3 = x − (−3): evaluate at −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

The remainder of p(x) ÷ (x + 1) is

A. −p(1)

B. p(−1)

C. p(0)

D. p(1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

573

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. p(−1)

GRADING NOTE x − (−1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $p(-3)$ for $p(x) = 2x^3 + 5x^2 - 4x - 3$.

$p(-3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

574

QUESTION ID
S16-B1-Q5

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 2

Factor the quadratic quotient • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

576

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $2x^2 - x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

577

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(x − 1)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

$x^2 + x - 6$ equals

A. $(x - 3)(x + 2)$

B. $(x - 6)(x + 1)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

578

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x + 3)(x − 2)

GRADING NOTE Product −6, sum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x^2 + 4$ factors over the integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

579

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

580

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 3)(x + 3)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 3

Divide a polynomial by a linear factor • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find q(x) if $x^3 - 3x^2 - 4x + 12 = (x - 2)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

582

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - x - 6$

GRADING NOTE Any equivalent form. Check by multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

Multiply $(x + 1)(x^2 + x - 6)$ to check a division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

583

QUESTION ID

S16-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 2x^2 - 5x - 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

True or false?

If $(x - 1)(x^2 + x - 6) = x^3 - 7x + 6$, then $x^2 + x - 6$ is the quotient of $x^3 - 7x + 6 \div (x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

584

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of the quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $x \neq 2$, $(x^3 - 8) \div (x - 2)$ equals

A. $x^2 + 2x + 4$

B. $x^2 + 4$

C. $x^2 - 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

585

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 2x + 4$

GRADING NOTE Difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The parentheses group the whole dividend; $x = 2$ is excluded because it makes the divisor 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find q(x) if $2x^3 + 5x^2 - 4x - 3 = (x + 3)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

586

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 - x - 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 1

Choose parameters so that a given binomial is a factor • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 3x + 9$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

588

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form. $27 + 9k - 9 + 9 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

Find a and b so that $x - 1$ and $x + 2$ are both factors of $x^3 + ax^2 + bx - 2$.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

589

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2; −1

GRADING NOTE Form required: both. $p(1) = 1 + a + b - 2 = 0$ and $p(-2) = -8 + 4a - 2b - 2 = 0$ give $a + b = 1$ and $2a - b = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

True or false?

If $x - 2$ and $x + 2$ are factors of $p(x)$, then $x^2 + 4$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

590

QUESTION ID

S16-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 2)(x + 2) = x^2 - 4$, not $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

For which k is x + 1 a factor of $x^4 + kx + 3$?

A. 2

B. −4

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

591

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $1 - k + 3 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

Write a cubic with leading coefficient 1 that has $x - 1$, $x + 1$, and $x - 4$ as factors, in standard form.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

592

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - 4x^2 - x + 4$

GRADING NOTE Form required: standard form. $(x^2 - 1)(x - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 2

Finish factorizations over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

$x - 1$ is a factor of $x^3 - 1$. Factor $x^3 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1/2 - (\sqrt{3}/2)i)(x + 1/2 + (\sqrt{3}/2)i)$

GRADING NOTE Any equivalent form. The quotient $x^2 + x + 1$ has zeros $-1/2 \pm (\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x + 2$ is a factor of $x^3 + 2x^2 + 9x + 18$. Factor it completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

595

QUESTION ID
S16-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 2)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

True or false?

Every factor found with the factor theorem gives a zero of the polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

596

QUESTION ID

S16-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x - a$ is a factor exactly when $p(a) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

$x - 2$ is a factor of $x^3 - 2x^2 + x - 2$. The other zeros are

A. i and −i

B. −2

C. 1 and −1

D. 2i and −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

597

QUESTION ID

S16-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. i and −i

GRADING NOTE Quotient $x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

$x - 3$ is a factor of $x^3 - 3x^2 + 4x - 12$. List all complex zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

598

QUESTION ID
S16-F2-Q5

TYPE
Short answer

CORRECT ANSWER
3, 2i, −2i

GRADING NOTE Form required: all three. Quotient $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

Identify zeros and multiplicities from a factorization

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Solve by the zero-product property	**HSA-REI.B.4.b**
BACKWARD 2	Factor to find hidden repeated factors	**HSA-SSE.A.2**
BACKWARD 3	An exponent counts repeated factors	**6.EE.A.1**
FORWARD 1	Build a polynomial from zeros and one more condition	**HSA-APR.B.3**
FORWARD 2	Decide what the zeros determine and what remains free	**HSA-APR.B.3**

SET 17 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

$p(x) = (x - 4)^2(x + 1)(2x - 3)$. List the zero of multiplicity 2, and the zeros of multiplicity 1.

Zero of multiplicity 2: _____

Zeros of multiplicity 1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

601

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

4; −1 or 3/2

GRADING NOTE Form required: both parts (3/2 or 1.5). $2x - 3 = 0$ gives $x = 3/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

$p(x) = x^3 - 6x^2 + 9x$ can be written as $x(x - a)^2$ with a positive number a. Find a, then give the zero of multiplicity 1 and the zero of multiplicity 2.

a = _____

Zero of multiplicity 1: _____

Zero of multiplicity 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

602

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

3; 0; 3

GRADING NOTE Form required: all three. $x^3 - 6x^2 + 9x = x(x^2 - 6x + 9) = x(x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

List the zeros of $p(x) = (x^2 - 9)(x^2 + 2x + 1)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

603

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 (1), −3 (1), −1 (2)

GRADING NOTE Form required: every zero with its multiplicity. $(x - 3)(x + 3)(x + 1)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

List the zeros of $p(x) = -2x(x + 5)^3$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

604

QUESTION ID

S17-M-Q4

TYPE

Short answer

CORRECT ANSWER

0 (1), −5 (3)

GRADING NOTE Form required: every zero with its multiplicity. The constant −2 is not a zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

Which gives the factorization of $p(x) = x^4 - 8x^2 + 16$ and its zeros with multiplicities?

A. $(x - 2)(x + 2)$; zeros 2 and -2, each of multiplicity 1

B. $(x^2 - 4)^2$; the only zero is 4, of multiplicity 2

C. $(x - 4)^2$; the only zero is 4, of multiplicity 2

D. $(x - 2)^2(x + 2)^2$; zeros 2 and -2, each of multiplicity 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

605

QUESTION ID

S17-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $(x - 2)^2(x + 2)^2$; zeros 2 and -2, each of multiplicity 2

GRADING NOTE $p = (x^2 - 4)^2 = ((x - 2)(x + 2))^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 1

Solve by the zero-product property • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 4)(x + 1) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

607

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 or −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x(2x − 3) = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

608

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0 or 3/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

True or false?

If $(x - 2)^2 = 0$, then $x = 2$ or $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

609

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $(x + 5)(x - 5) = 0$ are

A. 25

B. −5 and 5

C. 0 and 5

D. 5 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

610

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −5 and 5

GRADING NOTE Each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-2x(x + 5) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

611

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0 or −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 2

Factor to find hidden repeated factors • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 6x + 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

613

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

Factor $x^2 + 2x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

614

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^4 - 8x^2 + 16$ equals

A. $(x^2 + 4)^2$

B. $(x^2 - 4)^2$

C. $(x^2 - 8)^2$

D. $(x - 4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

615

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^2 - 4)^2$

GRADING NOTE $u^2 - 8u + 16$ with $u = x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x^3 - 6x^2 + 9x = x(x + 3)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

616

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x + 3)^2 = x^3 + 6x^2 + 9x$; the correct form is $x(x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

617

QUESTION ID

S17-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 3

An exponent counts repeated factors • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $(x - 4)^2$ as a product without an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

619

QUESTION ID

S17-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 4)(x − 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many factors of $(x + 5)$ are in $(x + 5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

620

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$(x - 2)^2(x - 2) = (x - 2)^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

621

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

In $7(x - 1)^4$, the factor $x - 1$ appears

A. 28 times

B. 4 times

C. 1 time

D. 7 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

622

QUESTION ID

S17-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4 times

GRADING NOTE Exponent 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (x + 1)(x + 1)(x − 3) with an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

623

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2(x - 3)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 1

Build a polynomial from zeros and one more condition • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write the cubic polynomial with zeros −1, 2, and 4 (each multiplicity 1) whose graph passes through (0, 16).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

625

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2(x + 1)(x − 2)(x − 4)

GRADING NOTE Any equivalent form. a(1)(−2)(−4) = 8a = 16 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

Write a polynomial of degree 3 with a zero of multiplicity 2 at x = 3, a zero at x = −2, and leading coefficient −1.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

626

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-(x-3)^2(x+2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

True or false?

Zeros alone determine a polynomial completely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

627

QUESTION ID

S17-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero constant multiple has the same zeros; a further condition fixes the scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

A quartic has zeros 1 and −1, each of multiplicity 2, and p(0) = 3. Then p(x) =

A. $-3(x-1)^2(x+1)^2$

B. $3(x^2-1)$

C. $(x-1)^2(x+1)^2$

D. $3(x-1)^2(x+1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

628

QUESTION ID

S17-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $3(x-1)^2(x+1)^2$

GRADING NOTE $a \cdot 1 \cdot 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

A polynomial has zeros 0 (multiplicity 2) and 5 (multiplicity 1), degree 3, and p(1) = −8. Find p(x).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

629

QUESTION ID

S17-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2(x - 5)$

GRADING NOTE Any equivalent form. a · 1 · (−4) = −8 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 2

Decide what the zeros determine and what remains free • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Which describes all polynomials of degree 2 whose only zeros are 1 and 3 (each of multiplicity 1)?

A. a(x + 1)(x + 3) for any real number a ≠ 0

B. a(x − 1)(x − 3) + c for any real numbers a and c

C. (x − 1)(x − 3) only

D. a(x − 1)(x − 3) for any real number a ≠ 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

631

QUESTION ID

S17-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. a(x − 1)(x − 3) for any real number a ≠ 0

GRADING NOTE The zeros fix the factors; a nonzero constant multiple keeps the zeros. The others drop the constant multiple, use the wrong signs, or add a constant (which moves the zeros).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

Which polynomial has 1 and 3 as its only real zeros but is NOT a constant multiple of $(x - 1)(x - 3)$?

A. $2(x - 1)(x - 3)$

B. $(x - 1)(x - 3)(x + 1)$

C. $(x - 1)^2(x - 3)(x^2 + 1)$

D. $(x - 1)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

632

QUESTION ID

S17-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 1)^2(x - 3)(x^2 + 1)$

GRADING NOTE $x^2 + 1$ has no real zeros, and the extra factor $(x - 1)$ only repeats a zero. The others are a constant multiple, add the real zero -1, or have the zero -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

True or false?

Two cubic polynomials with the same three zeros (each multiplicity 1) must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

633

QUESTION ID

S17-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They can differ by a constant factor, for example $(x-1)(x-2)(x-3)$ and $2(x-1)(x-2)(x-3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

Which extra information fixes a cubic with zeros −2, 0, 2?

A. That it has degree 3

B. That 0 is a zero

C. Its end behavior only

D. Its value at x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

634

QUESTION ID

S17-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Its value at x = 1

GRADING NOTE One point fixes the scale a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

Why may the real zeros of a degree-4 polynomial fail to determine it, even up to a constant multiple?

A. Every degree-4 polynomial with real coefficients has four real zeros counted with multiplicity, so its real zeros always determine it up to a constant multiple.

B. Two polynomials with the same real zeros must be equal.

C. It may have non-real zeros, from a factor such as $x^2 + c$ with $c > 0$, that the real zeros do not show; for example $(x - 1)(x - 2)(x^2 + 1)$ and $(x - 1)(x - 2)(x^2 + 4)$ have the same real zeros.

D. They never fail: the real zeros of a degree-4 polynomial determine it exactly, including its leading coefficient, so $(x - 1)(x - 2)$ gives the whole polynomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

635

QUESTION ID

S17-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. It may have non-real zeros, from a factor such as $x^2 + c$ with $c > 0$, that the real zeros do not show; for example $(x - 1)(x - 2)(x^2 + 1)$ and $(x - 1)(x - 2)(x^2 + 4)$ have the same real zeros.

GRADING NOTE Non-real zeros are invisible on the real line. The others make false claims ($x^4 + 1$ has no real zeros; multiples share zeros).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

Identify the rough graph of a polynomial from its zeros

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Read zeros and multiplicities	**HSA-APR.B.3**
BACKWARD 2	End behavior from the leading term	**HSF-IF.B.4**
BACKWARD 3	Find the y-intercept by evaluating at 0	**HSF-IF.A.2**
BACKWARD 4	Sign of a power of a negative or positive number	**7.NS.A.2**
FORWARD 1	Sketch graphs from standard form by factoring first	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its rough graph	**HSA-APR.B.3**

SET 18 • HSA-APR.B.3

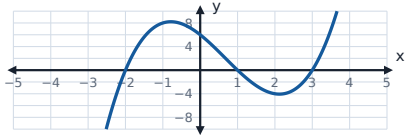
MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

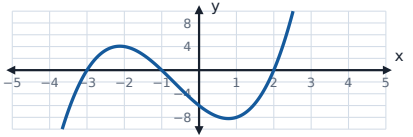
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = (x + 2)(x - 1)(x - 3)$?

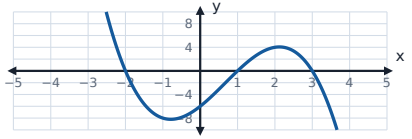
A



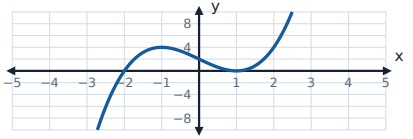
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

638

QUESTION ID

S18-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Zeros −2, 1, 3 (each crossed, multiplicity 1); y-intercept (2)(−1)(−3) = 6; degree 3 with positive leading coefficient, so it falls left and rises right. The others have the zeros with the wrong signs, the opposite end behavior, or a double zero at 1.

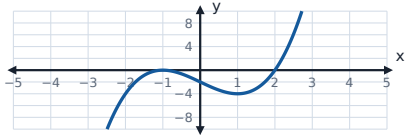
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

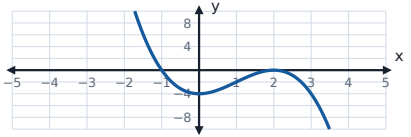
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = -(x + 1)^2(x - 2)$?

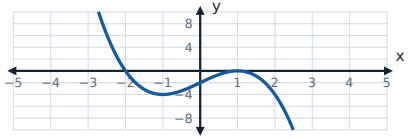
A



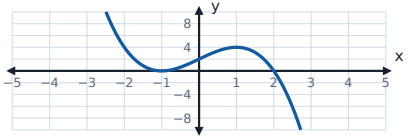
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

639

QUESTION ID

S18-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Double zero at −1 (touches), single zero at 2 (crosses); y-intercept $-(1)^2(-2) = 2$; leading coefficient −1 with odd degree, so it rises left and falls right. The others flip the sign, swap the multiplicities, or use the zeros 1 and −2.

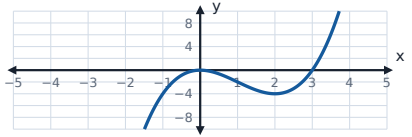
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

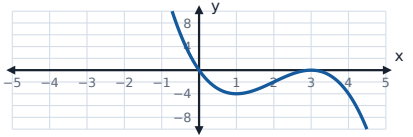
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = x(x - 3)^2$?

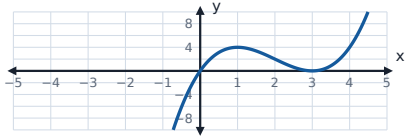
A



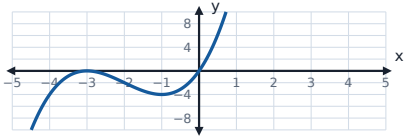
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

640

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Single zero at 0 (crosses), double zero at 3 (touches); y-intercept 0; $y(1) = 4 > 0$; falls left and rises right. The others swap the multiplicities, flip the sign, or put the double zero at -3.

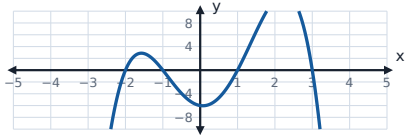
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

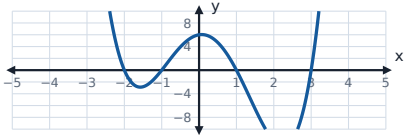
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = (x + 2)(x + 1)(x - 1)(x - 3)$?

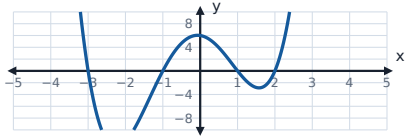
A



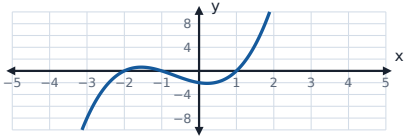
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

641

QUESTION ID

S18-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

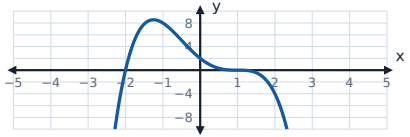
GRADING NOTE Four single zeros −2, −1, 1, 3; y-intercept (2)(1)(−1)(−3) = 6; degree 4 with positive leading coefficient, so it rises at both ends. The others are reflected, have zeros with the wrong signs (same y-intercept), or leave out a factor.

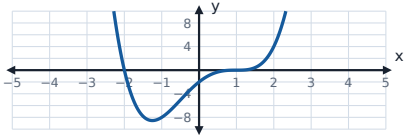
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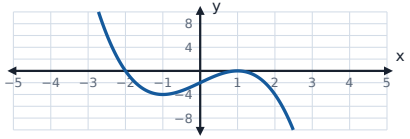
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

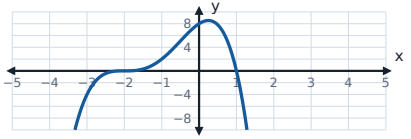
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = -(x - 1)^3(x + 2)$?

A 

B 

C 

D 

A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

642

QUESTION ID

S18-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Zero 1 has multiplicity 3 (crosses with a flattened shape), zero −2 is single; y-intercept $-(-1)^3(2) = 2$; degree 4 with negative leading coefficient, so it falls at both ends. The others flip the sign, use multiplicity 2 at 1, or put the triple zero at −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 1

Read zeros and multiplicities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

644

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$-2, 1, 3$

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

List the zeros of $-(x + 1)^2(x - 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

645

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−1 (2), 2 (1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

True or false?

$x(x - 3)^2$ has a zero of multiplicity 2 at $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

646

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 0$ has multiplicity 1; $x = 3$ has multiplicity 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

The zero $x = 1$ of $-(x - 1)^3(x + 2)$ has multiplicity

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

647

QUESTION ID

S18-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Exponent 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x + 1)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

QUESTION ID

S18-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2, −1, 1, 3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 2

End behavior from the leading term • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

As x grows large and positive, what happens to x^3?

A. It grows large and negative.

B. It grows large and positive.

C. It approaches 1.

D. It approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

650

QUESTION ID

S18-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. It grows large and positive.

GRADING NOTE A positive number cubed is positive and large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

As x grows large and negative, what happens to $-x^4$?

A. It becomes large and negative.

B. It approaches −1.

C. It becomes large and positive.

D. It approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

651

QUESTION ID

S18-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. It becomes large and negative.

GRADING NOTE x^4 is large and positive, so $-x^4$ is large and negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

The leading term of $(x + 2)(x - 1)(x - 3)$ is

A. $-x^3$

B. -6

C. $3x$

D. x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

652

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^3

GRADING NOTE $x \cdot x \cdot x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

True or false?

A polynomial of even degree with a positive leading coefficient falls at both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

653

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Like x^2, it rises at both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the leading term of $-(x-1)^3(x+2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

654

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-x^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 3

Find the y-intercept by evaluating at 0 • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find y when x = 0 for y = (x + 2)(x − 1)(x − 3).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

656

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 2 · (−1) · (−3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

Find y when x = 0 for $y = -(x + 1)^2(x - 2)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

657

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. −(1)(−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

True or false?

The y-intercept of $x(x - 3)^2$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

658

QUESTION ID

S18-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 0 (a factor x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

The y-intercept of $-(x-1)^3(x+2)$ is

A. 0

B. -2

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

659

QUESTION ID

S18-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $-(-1)(2) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

Find y when x = 1 for $y = x(x - 3)^2$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

660

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 4

Sign of a power of a negative or positive number • Grade 7 • The Number System • 7.NS.A.2

Is $(-0.5)^2$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

662

QUESTION ID

S18-B4-Q1

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

Is $(-0.5)^3$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

663

QUESTION ID

S18-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

True or false?

An even power of a negative number is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

664

QUESTION ID

S18-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs of negative factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

Near $x = 3$, $(x - 3)^2$ is

A. negative on both sides

B. positive on both sides

C. zero everywhere

D. positive on the left and negative on the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

665

QUESTION ID

S18-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. positive on both sides

GRADING NOTE A square does not change sign at $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

Does $(x - 1)^3$ change sign at $x = 1$?

A. No: it is negative on both sides of $x = 1$.

B. No: it is positive on both sides of $x = 1$.

C. Yes: it is positive for $x < 1$ and negative for $x > 1$.

D. Yes: it is negative for $x < 1$ and positive for $x > 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

666

QUESTION ID

S18-B4-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Yes: it is negative for x < 1 and positive for x > 1.

GRADING NOTE An odd power keeps the sign of $x - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

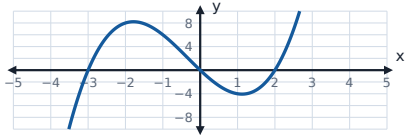
SET 18 • HSA-APR.B.3

FORWARD 1

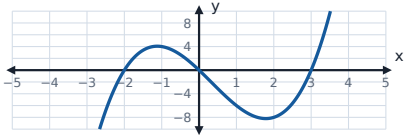
Sketch graphs from standard form by factoring first • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Factor $y = x^3 - x^2 - 6x$. Which graph is a rough graph of it?

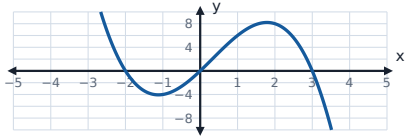
A



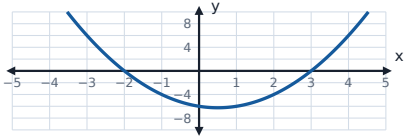
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

668

QUESTION ID

S18-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE $x^3 - x^2 - 6x = x(x - 3)(x + 2)$: crosses at -2, 0, 3; y-intercept 0; falls left and rises right. The others use $x(x + 3)(x - 2)$, the reflection, or drop the factor x.

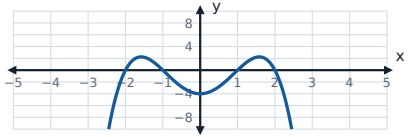
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

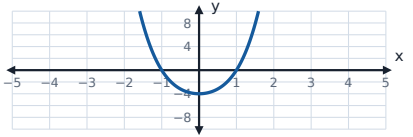
ANSWER KEY Reduced copy of the question page, with its answer

Factor $y = x^4 - 5x^2 + 4$. Which graph is a rough graph of it?

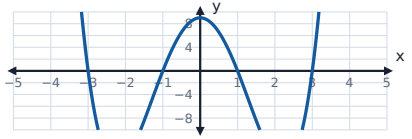
A



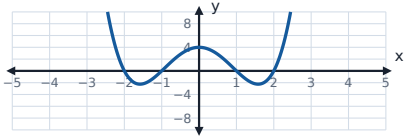
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

669

QUESTION ID

S18-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$: crosses at ±1 and ±2; y-intercept 4; rises at both ends. The others are reflected, factor as $(x^2 - 1)(x^2 + 4)$, or use $(x^2 - 1)(x^2 - 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

True or false?

$y = x^3 + 4x$ crosses the x-axis at $x = -2$, 0, and 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

670

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x^2 + 4) = 0$ only at $x = 0$; $x^2 + 4$ has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

$y = -x^3 + 9x$ crosses the x-axis at

A. 0 and 9

B. 3 only

C. −3, 0, 3

D. −9, 0, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

671

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −3, 0, 3

GRADING NOTE $-x(x - 3)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

$y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$. What does the graph do at each zero?

A. It crosses the x-axis at $x = 1$ and touches it at $x = -2$.

B. It crosses the x-axis at $x = -1$ and touches it (turns back) at $x = 2$.

C. It crosses the x-axis at both $x = -1$ and $x = 2$.

D. It touches the x-axis at $x = -1$ and crosses it at $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

672

QUESTION ID

S18-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. It crosses the x-axis at $x = -1$ and touches it (turns back) at $x = 2$.

GRADING NOTE Odd multiplicity: crosses; even multiplicity: touches. The others swap the multiplicities or the signs of the zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

FORWARD 2

Write a polynomial from its rough graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A cubic's graph crosses the x-axis at −3 and touches it at 2, and passes through (0, 12). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

674

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 2)^2$

GRADING NOTE Any equivalent form. a(3)(4) = 12 gives a = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

A quartic rises at both ends, touches the x-axis at −1 and at 3, and passes through (0, 18). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

675

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2(x + 1)^2(x - 3)^2$

GRADING NOTE Any equivalent form. a(1)(9) = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

True or false?

A graph that touches the x-axis at x = a without crossing has a zero of even multiplicity at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

676

QUESTION ID

S18-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The factor does not change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

A cubic falls to the right and crosses the x-axis at 0, 1, and 4. Its leading coefficient is

A. zero

B. not determined by the end behavior

C. positive

D. negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

677

QUESTION ID

S18-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. negative

GRADING NOTE A cubic that falls to the right has a negative leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

A polynomial graph crosses at −2, touches at 1, falls to the left and rises to the right, and has y-intercept 4. Give a polynomial of least degree.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

678

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2(x + 2)(x - 1)^2$

GRADING NOTE Any equivalent form. Degree 3 with a > 0; a(2)(1) = 4 gives a = 2, which agrees with the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

Prove polynomial identities

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Expand the square of a binomial	**HSA-APR.A.1**
BACKWARD 2	Recognize a difference of squares	**HSA-SSE.A.2**
BACKWARD 3	What makes an identity	**6.EE.A.4**
FORWARD 1	Prove or disprove proposed identities	**HSA-APR.C.4**
FORWARD 2	Prove identities with the Binomial Theorem	**HSA-APR.C.5**

SET 19 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$?

A. For $x = 2$ and $y = 1$: $25 = 9 + 16$, so the identity is true.

B. Take the square root of both sides: $x^2 + y^2 = (x^2 - y^2) + 2xy$, and the right side simplifies to $x^2 - y^2 + 2xy$, so the identity is true.

C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

D. The right side is $(x^2 - y^2)^2 + (2xy)^2 = x^4 - y^4 + 4x^2y^2$, and the left side is $x^4 + y^4 + 4x^2y^2$, so the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

681

QUESTION ID
S19-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

GRADING NOTE Both sides expand to $x^4 + 2x^2y^2 + y^4$. The others check one example, square $x^2 - y^2$ as $x^4 - y^4$, or take square roots term by term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

Which argument proves the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$?

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

B. For $a = 1$ and $b = 2$ the left side is $1 + 8 = 9$ and the right side is $3 \cdot (1 - 2 + 4) = 9$, so the identity is true for all a and b.

C. Multiply the first terms and the last terms: $a \cdot a^2 = a^3$ and $b \cdot b^2 = b^3$, so the product is $a^3 + b^3$.

D. The product is $a^3 - 2a^2b + b^3$, which equals $a^3 + b^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

682

QUESTION ID

S19-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

GRADING NOTE All six products, then the cancellations. The others check one example, skip four of the six products, or multiply incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

Which argument proves the identity $(a + b)^2 - (a - b)^2 = 4ab$?

A. For $a = 3$ and $b = 1$: $16 - 4 = 12 = 4 \cdot 3$, so the identity is true.

B. $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab$.

C. $(a + b)^2 - (a - b)^2 = [(a + b) - (a - b)]^2 = (2b)^2 = 4ab$.

D. $(a + b)^2 - (a - b)^2 = (a^2 + b^2) - (a^2 - b^2) = a^2 + b^2 - a^2 + b^2 = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

683

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab.$

GRADING NOTE Subtract the whole second square, changing the sign of every term (or factor as $[(a + b) - (a - b)][(a + b) + (a - b)] = (2b)(2a)$). The others check one example, square binomials as $a^2 + b^2$, or use the false rule $x^2 - y^2 = (x - y)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

Which argument proves the identity $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$?

A. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 + (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

B. For $x = 0$: $4 = 2 \cdot 2$, so the identity is true.

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

D. Multiply the first terms and the last terms: $x^2 \cdot x^2 = x^4$ and $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

684

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

GRADING NOTE Group as a difference of squares (or expand all nine products). The others check one example, use $(u + v)(u - v) = u^2 + v^2$, or skip the middle products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

Which argument proves the identity
$(n + 1)^3 - n^3 = 3n^2 + 3n + 1$?

A. For n = 1, 2, and 3 the left side is 7, 19, and 37, and so is the right side, so it is true for every n.

B. $(n + 1)^3 = n^3 + 1$, so $(n + 1)^3 - n^3 = 1$, which is $3n^2 + 3n + 1$.

C. $(n + 1)^3 - n^3 = (n + 1 - n)^3 = 1^3 = 1$.

D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

685

QUESTION ID
S19-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

GRADING NOTE Expand $(n + 1)^3$ correctly and subtract. The others check several values (evidence, not a proof) or use the false rules $a^3 - b^3 = (a - b)^3$ and $(n + 1)^3 = n^3 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 1

Expand the square of a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

687

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$a^2 + 2a \cdot b + b^2$

GRADING NOTE Any equivalent expanded form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x^2 - y^2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

688

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^4 - 2x^2y^2 + y^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

True or false?

$(a - b)^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

689

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(a - b)^2 = a^2 - 2ab + b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2xy)^2$ equals

A. $2xy^2$

B. $4xy$

C. $2x^2y^2$

D. $4x^2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

690

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $4x^2y^2$

GRADING NOTE Square each factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

Expand $(n + 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

691

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$n^3 + 3n^2 + 3n + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 2

Recognize a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $a^2 - b^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

693

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(a − b)(a + b)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(x^2 + 2)^2 - (2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

694

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 2 - 2x)(x^2 + 2 + 2x)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

True or false?

$u^2 - v^2 = (u - v)(u + v)$ for any expressions u and v.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

695

QUESTION ID

S19-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Structure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

Which is a difference of squares?

A. $x^3 - 1$

B. $x^2 - 3x$

C. $9x^2 - 16$

D. $9x^2 + 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

696

QUESTION ID

S19-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $9x^2 - 16$

GRADING NOTE $(3x)^2 - 4^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

Factor $25 - x^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

697

QUESTION ID

S19-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5 - x^2)(5 + x^2)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 3

What makes an identity • Grade 6 • Expressions & Equations • 6.EE.A.4

Test $(a + b)^2 = a^2 + b^2$ with $a = 1$ and $b = 2$. Which conclusion is correct?

A. It is not an identity: the left side is 9 and the right side is 5.

B. The test is inconclusive: one example can never show that a statement is not an identity.

C. It is an identity: both sides equal 9.

D. It is an identity: both sides equal 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

699

QUESTION ID

S19-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. It is not an identity: the left side is 9 and the right side is 5.

GRADING NOTE $(1 + 2)^2 = 9$ and $1^2 + 2^2 = 5$; one counterexample shows it is not an identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

Which statement describes the difference between an identity and an equation?

A. An identity is true for at least one example; an equation is true for no values.

B. An identity has no variables; an equation has variables.

C. An identity is true for every value of its variables; an equation may be true only for some values.

D. An identity is an equation that has exactly one solution, found by solving it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

700

QUESTION ID

S19-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. An identity is true for every value of its variables; an equation may be true only for some values.

GRADING NOTE For example $(x + 1)^2 = x^2 + 2x + 1$ is an identity, while $x + 1 = 3$ is true only for $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

True or false?

Checking an identity with three numbers proves it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

701

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples cannot prove a statement about all values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

$2(x + 3) = 2x + 6$ is

A. an identity

B. true only for $x = 3$

C. true only for $x = 0$

D. never true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

702

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. an identity

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

Which values show that $a^3 + b^3 = (a + b)^3$ is NOT an identity?

A. a = 0, b = 2 (left side 8, right side 8)

B. a = 1, b = −1 (left side 0, right side 0)

C. a = 1, b = 1 (left side 2, right side 8)

D. a = 2, b = 0 (left side 8, right side 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

703

QUESTION ID

S19-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a = 1, b = 1 (left side 2, right side 8)

GRADING NOTE A counterexample must make the two sides different; only a = b = 1 does among these choices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 1

Prove or disprove proposed identities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Is $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ an identity? Which answer is correct?

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

B. Yes, because $a = 1$ and $b = 1$ give $8 = 2 + 6$.

C. No: for $a = 1$ and $b = 2$ the left side is $3^3 = 27$ and the right side is $1 + 8 + 3 \cdot 2 = 15$, so the sides differ.

D. No: $(a + b)^3 = a^3 + b^3$, so the extra term makes the two sides different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

705

QUESTION ID

S19-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

GRADING NOTE True by expansion. The others use $(a + b)^3 = a^3 + b^3$, check one example, or miscompute ($a = 1$, $b = 2$ gives $27 = 1 + 8 + 18$ on both sides).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

Is $a^4 - b^4 = (a - b)^4$ an identity? Which answer is correct?

A. Yes: for a = 1 and b = 0 both sides equal 1.

B. No: for a = 2 and b = 1, the left side is 16 − 1 = 15 and the right side is $1^4 = 1$.

C. Yes: a fourth power applies to each term inside the parentheses, so $(a - b)^4 = a^4 - b^4$ for all a and b.

D. No, because $a^4 - b^4 = (a - b)^2(a + b)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

706

QUESTION ID

S19-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No: for a = 2 and b = 1, the left side is 16 − 1 = 15 and the right side is $1^4 = 1$.

GRADING NOTE One counterexample disproves an identity. The others distribute the exponent, generalize from one example, or use a false factorization ($a^4 - b^4 = (a - b)(a + b)(a^2 + b^2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

True or false?

$(x - 1)(x^2 + x + 1) = x^3 + 1$ is an identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

707

QUESTION ID

S19-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The product is $x^3 - 1$ (difference of cubes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

Which is an identity?

A. $(x + 2)^2 = x^2 + 4$

B. $(x + 1)^3 = x^3 + 1$

C. $x^2 - 4 = (x - 4)^2$

D. $(x + 2)^2 - (x - 2)^2 = 8x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

708

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $(x + 2)^2 - (x - 2)^2 = 8x$

GRADING NOTE 4ab with a = x, b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $(x + c)^2 - (x - c)^2 = 12x$ is an identity.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

709

QUESTION ID

S19-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 4cx = 12x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 2

Prove identities with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Which argument uses the Binomial Theorem to prove that the entries in row n of Pascal's Triangle add to 2^n?

A. Row n has n + 1 entries, and each entry is the sum of the two entries above it, so every row has twice as many entries as the row before it and the sum of row n is 2^n.

B. Row 3 is 1, 3, 3, 1, which adds to $8 = 2^3$, so every row n adds to 2^n.

C. Set x = y = 1 in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + \ldots + C(n, n)y^n$. The left side is 2^n and the right side is C(n, 0) + C(n, 1) + ... + C(n, n), the row sum.

D. Set x = 1 and y = 0 in $(x + y)^n = C(n, 0)x^n + \ldots + C(n, n)y^n$: the left side is 1 and the right side is C(n, 0) = 1, so each row adds to 2^n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

711

QUESTION ID
S19-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Set x = y = 1 in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + \ldots + C(n, n)y^n$. The left side is 2^n and the right side is C(n, 0) + C(n, 1) + ... + C(n, n), the row sum.

GRADING NOTE Substituting 1 for x and y makes every term equal to its coefficient. The others check one row, use a substitution that gives 1, or argue from the number of entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

Use the Binomial Theorem with $x = 1$ and $y = -1$ to find $C(5,0) - C(5,1) + C(5,2) - C(5,3) + C(5,4) - C(5,5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

712

QUESTION ID
S19-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. $(1 - 1)^5 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

True or false?

$(1 + 1)^4 = 1 + 4 + 6 + 4 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

713

QUESTION ID
S19-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

The sum of the coefficients of $(2x + 1)^3$ is

A. 9

B. 8

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

714

QUESTION ID
S19-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 27

GRADING NOTE Set x = 1: 3^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $(1 + 0.1)^3 = 1 + 3(0.1) + 3(0.01) + 0.001$ to find 1.1^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

715

QUESTION ID
S19-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1.331

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

Use polynomial identities to describe numerical relationships

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Substitute values into an identity	**6.EE.A.2**
BACKWARD 2	Check a Pythagorean triple	**8.G.B.6**
BACKWARD 3	Choose a and b to fit a product	**HSA-APR.C.4**
FORWARD 1	Generate a triple with a required side	**HSA-APR.C.4**
FORWARD 2	Connect the triple identity to complex numbers	**HSN-CN.A.2**

SET 20 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ generates Pythagorean triples. Use $x = 3$ and $y = 2$ to find a triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

718

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

5, 12, 13

GRADING NOTE Any order. $x^2 - y^2 = 5$, $2xy = 12$, $x^2 + y^2 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

Use the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ with $x = 5$ and $y = 2$ to find a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

719

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

20, 21, 29

GRADING NOTE Any order. $25 - 4 = 21$, $2 \cdot 5 \cdot 2 = 20$, $25 + 4 = 29$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

Use the identity $(a + b)(a - b) = a^2 - b^2$ to compute 47×53 mentally: write $53 = a + b$ and $47 = a - b$ with whole numbers a and b.

a = _____

b = _____

47 × 53 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

720

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

50; 3; 2491

GRADING NOTE Form required: all three. a = 50 and b = 3; $50^2 - 3^2 = 2500 - 9 = 2491$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

Use the identity $(n + 1)^2 - n^2 = 2n + 1$ to write 41 as $(n + 1)^2 - n^2$, a difference of two consecutive squares. Give $n + 1$ and n.

$n + 1 =$ _____

$n =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

721

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

21; 20

GRADING NOTE Form required: both. $2n + 1 = 41$ gives $n = 20$; $21^2 - 20^2 = 441 - 400 = 41$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

Use the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$ to compute $13^3 - 12^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

722

QUESTION ID
S20-M-Q5

TYPE
Short answer

CORRECT ANSWER
469

GRADING NOTE Any equivalent form. 3·144 + 36 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 1

Substitute values into an identity • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 - y^2$ for $x = 3$ and $y = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

724

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2xy for x = 5 and y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

725

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

True or false?

For $n = 12$, $3n^2 + 3n + 1 = 433$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

726

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $432 + 36 + 1 = 469$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $a = 50$ and $b = 3$, $a^2 - b^2$ is

A. 2491

B. 2500

C. 2494

D. 2509

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

727

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2491

GRADING NOTE 2500 − 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2n + 1 for n = 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

728

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

41

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 2

Check a Pythagorean triple • Grade 8 • Geometry • 8.G.B.6

Do 5, 12, 13 satisfy $a^2 + b^2 = c^2$?

A. No: 25 + 144 = 196

B. Yes: 25 + 144 = 169

C. No: 25 + 144 ≠ 169

D. Yes: 5 + 12 = 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

730

QUESTION ID

S20-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Yes:

25 + 144 = 169

GRADING NOTE $5^2 + 12^2 = 25 + 144 = 169 = 13^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

Do 20, 21, 29 satisfy $a^2 + b^2 = c^2$?

A. Yes: 400 + 441 = 841

B. No: 400 + 441 = 851

C. No: 20 + 21 ≠ 29

D. Yes: 20 + 21 = 41

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

731

QUESTION ID

S20-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Yes:

400 + 441 = 841

GRADING NOTE $20^2 + 21^2 = 400 + 441 = 841 = 29^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

True or false?

6, 8, 11 is a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

732

QUESTION ID

S20-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $36 + 64 = 100 \neq 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

Which is a Pythagorean triple?

A. 8, 15, 17

B. 6, 7, 9

C. 2, 3, 4

D. 4, 5, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

733

QUESTION ID

S20-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

GRADING NOTE 64 + 225 = 289.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

Why is a triangle with sides 5, 12, 13 a right triangle?

A. 13 is the longest side, so the angle opposite it must be 90° in any triangle.

B. 5 + 12 > 13 satisfies the triangle inequality, so it is a right triangle.

C. Its three sides are whole numbers, so it must be a right triangle.

D. $5^2 + 12^2 = 13^2$, so by the converse of the Pythagorean Theorem it is a right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

734

QUESTION ID

S20-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $5^2 + 12^2 = 13^2$, so by the converse of the Pythagorean Theorem it is a right triangle.

GRADING NOTE The converse of the Pythagorean Theorem. The others give the triangle inequality (true for every triangle) or non-reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 3

Choose a and b to fit a product • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 47 and 53 as 50 − b and 50 + b. What is b?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

736

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

Write 98 × 102 as (a − b)(a + b).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

737

QUESTION ID

S20-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(100 − 2)(100 + 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$29 \times 31 = 30^2 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

738

QUESTION ID

S20-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(30 - 1)(30 + 1) = 30^2 - 1 = 899.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

64 × 56 equals

A. $(60 - 4)^2$

B. $64^2 - 56^2$

C. $60^2 - 4^2$

D. $60^2 + 4^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

739

QUESTION ID

S20-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $60^2 - 4^2$

GRADING NOTE a = 60, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 98 × 102.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

740

QUESTION ID

S20-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9996

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 1

Generate a triple with a required side • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ gives triples with legs $x^2 - y^2$ and $2xy$ and hypotenuse $x^2 + y^2$. Find whole numbers $x > y$ that give the triple with legs 7 and 24 and hypotenuse 25.

x = _____

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

742

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4; 3

GRADING NOTE Form required: both. $x^2 + y^2 = 25$, $x^2 - y^2 = 7$, $2xy = 24$ give $x = 4$, $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ gives triples with legs $x^2 - y^2$ and $2xy$ and hypotenuse $x^2 + y^2$. Find whole numbers $x > y$ that give a leg $2xy = 40$ and hypotenuse 41.

x = _____

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

743

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5; 4

GRADING NOTE Form required: both. $xy = 20$ and $x^2 + y^2 = 41$ give $x = 5$, $y = 4$ (the other leg is 9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

True or false?

Every choice of whole numbers $x > y > 0$ in the identity gives a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

744

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All three expressions are positive whole numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

Which choice gives the triple 8, 15, 17?

A. x = 4, y = 1

B. x = 5, y = 3

C. x = 4, y = 2

D. x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

745

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x = 4, y = 1

GRADING NOTE 16 − 1 = 15, 8, 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

The identity gives legs $x^2 - y^2$ and $2xy$. Why does $x = 2y$ (y a positive whole number) give legs in the ratio 3 : 4?

A. With y = 1, x = 2 gives legs 3 and 4, so every y works the same way without checking.

B. $x^2 - y^2 = 2y^2$ and $2xy = 4y^2$, so the ratio is 1 : 2, which equals 3 : 4.

C. $x^2 - y^2 = 3y$ and $2xy = 4y$, so the legs are 3y and 4y.

D. $x^2 - y^2 = 4y^2 - y^2 = 3y^2$ and $2xy = 4y^2$, so the legs are $3y^2$ and $4y^2$, in the ratio 3 : 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

746

QUESTION ID

S20-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - y^2 = 4y^2 - y^2 = 3y^2$ and $2xy = 4y^2$, so the legs are $3y^2$ and $4y^2$, in the ratio 3 : 4.

GRADING NOTE Substitute x = 2y. The others compute $(2y)^2$ incorrectly, generalize from one case, or drop the squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 2

Connect the triple identity to complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(3 + 2i)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

748

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 12i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

Compute $(x + yi)^2$ in the form $a + bi$, where x and y are real. Give the real part and the imaginary part.

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

749

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - y^2$; $2 \cdot x \cdot y$

GRADING NOTE Form required: both. $(x + yi)^2 = x^2 + 2xyi + y^2i^2 = (x^2 - y^2) + 2xyi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

True or false?

For positive integers $x > y$, the real and imaginary parts of $(x + yi)^2$ are $x^2 - y^2$ and $2xy$, the legs of a Pythagorean triple whose hypotenuse is $x^2 + y^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

750

QUESTION ID

S20-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + yi)^2 = (x^2 - y^2) + 2xyi$, and $(x^2 - y^2)^2 + (2xy)^2 = (x^2 + y^2)^2$. The condition $x > y > 0$ makes both legs positive integers ($x = y = 1$ would give a zero leg).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

$(5 + 2i)^2$ equals

A. 25 + 4i

B. 21 + 20i

C. 21 + 10i

D. 29 + 20i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

751

QUESTION ID

S20-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 21 + 20i

GRADING NOTE 25 − 4 + 20i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

Use $(4 + i)^2$ to find a Pythagorean triple.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

752

QUESTION ID

S20-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15, 8, 17

GRADING NOTE Any order. $(4 + i)^2 = 15 + 8i$; $4^2 + 1^2 = 17$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

Expand with the Binomial Theorem

MAIN	Algebra II main question	**HSA-APR.C.5**
BACKWARD 1	Multiply binomials repeatedly	**HSA-APR.A.1**
BACKWARD 2	Build Pascal's Triangle by a rule	**4.OA.C.5**
BACKWARD 3	Powers of products and of negatives	**8.EE.A.1**
FORWARD 1	Find a specific term or coefficient with a condition	**HSA-APR.C.5**
FORWARD 2	Use binomial coefficients to compute probabilities	**HSS-CP.B.9**

SET 21 • HSA-APR.C.5

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem to expand $(x + 2)^5$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

755

QUESTION ID
S21-M-Q1

TYPE
Short answer

CORRECT ANSWER
$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

GRADING NOTE Form required: standard form. Coefficients $C(5, k)\cdot 2^k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

Use the Binomial Theorem to expand $(2a - b)^4$. Write every term multiplied out.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

756

QUESTION ID
S21-M-Q2

TYPE
Short answer

CORRECT ANSWER
$16a^4 - 32a^3b + 24a^2b^2 - 8a \cdot b^3 + b^4$

GRADING NOTE Form required: expanded (any order of terms). Coefficients 1, 4, 6, 4, 1 with powers of 2a and −b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

Use the Binomial Theorem to find the coefficient of x^3 in $(x + 3)^7$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

757

QUESTION ID
S21-M-Q3

TYPE
Short answer

CORRECT ANSWER
2835

GRADING NOTE Any equivalent form. $C(7, 3)\cdot 3^4 = 35\cdot 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

Use the Binomial Theorem to find the term containing x^3y^3 in $(x - 2y)^6$.

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

758

QUESTION ID
S21-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-160x^3y^3$

GRADING NOTE Any equivalent form. $C(6, 3)(-2)^3 = 20\cdot(-8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q4**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

Use the first three terms of the binomial expansion of $(1 + 0.02)^5$ to estimate 1.02^5 to four decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

759

QUESTION ID
S21-M-Q5

TYPE
Short answer

CORRECT ANSWER
1.1040

GRADING NOTE Any value 1.1040 or 1.1041 is accepted (1 + 0.1 + 0.004 = 1.104; the exact value is 1.10408...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 1

Multiply binomials repeatedly • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

761

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 4x + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

762

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 6x^2 + 12x + 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$(x + y)^3 = x^3 + y^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

763

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$; for $x = y = 1$, $8 \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The number of terms in the expansion of $(x + y)^5$ is

A. 6

B. 5

C. 10

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

764

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE n + 1 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

Expand $(a - b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

765

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - 2a \cdot b + b^2$

GRADING NOTE Any equivalent expanded form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 2

Build Pascal's Triangle by a rule • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Row 4 of Pascal's Triangle is 1, 4, 6, 4, 1. Write row 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

767

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1, 5, 10, 10, 5, 1

GRADING NOTE Form required: all six entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

Write row 7 of Pascal's Triangle (row 6 is 1, 6, 15, 20, 15, 6, 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

768

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 7, 21, 35, 35, 21, 7, 1

GRADING NOTE Form required: all entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

True or false?

Each inner entry of Pascal's Triangle is the sum of the two entries above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

769

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rule of the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

The fourth entry of row 7 (counting from the first 1) is

A. 20

B. 7

C. 35

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

770

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 35

GRADING NOTE 1, 7, 21, 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the sum of the entries in row 5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

771

QUESTION ID

S21-B2-Q5

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 3

Powers of products and of negatives • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(2a)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

773

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$16a^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-2y)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

774

QUESTION ID

S21-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$-8y^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$(-b)^4 = -b^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

775

QUESTION ID

S21-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An even power of $-b$ is positive: $(-b)^4 = b^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

3^4 equals

A. 27

B. 12

C. 64

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

776

QUESTION ID

S21-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 81

GRADING NOTE 3·3·3·3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(0.02)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

777

QUESTION ID

S21-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.0004

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 1

Find a specific term or coefficient with a condition • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Find the constant term of $(x + 1/x)^6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

779

QUESTION ID
S21-F1-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE Any equivalent form. Middle term C(6, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

The coefficient of x^2 in $(1 + kx)^5$ is 90. Find k ($k > 0$).

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

780

QUESTION ID
S21-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. $C(5, 2)k^2 = 10k^2 = 90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

True or false?

The middle term of $(x + y)^8$ is $56x^4y^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

781

QUESTION ID

S21-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $C(8, 4) = 70$, so the middle term is $70x^4y^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

The term with x^4 in $(x^2 - 1)^5$ is

A. $-5x^4$

B. $-10x^4$

C. $10x^4$

D. $5x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

782

QUESTION ID
S21-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $-10x^4$

GRADING NOTE $C(5, 2)(x^2)^2(-1)^3 = -10x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

Find the coefficient of x in $(2x - 1)^6$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

783

QUESTION ID
S21-F1-Q5

TYPE
Short answer

CORRECT ANSWER
−12

GRADING NOTE Any equivalent form. $C(6, 1)\cdot 2\cdot(-1)^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 2

Use binomial coefficients to compute probabilities • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A fair coin is tossed 5 times. Use C(5, k) to find P(exactly 3 heads). Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

785

QUESTION ID
S21-F2-Q1

TYPE
Short answer

CORRECT ANSWER
5/16

GRADING NOTE $C(5, 3)/2^5 = 10/32 = 5/16$ (0.3125).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner lands on red with probability 0.3. Find P(exactly 2 reds in 4 spins).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

786

QUESTION ID
S21-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.2646

GRADING NOTE Any equivalent form. $C(4, 2)(0.3)^2(0.7)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

True or false?

The probabilities of 0, 1, ..., n successes add to 1 because $(p + (1 - p))^n = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

787

QUESTION ID
S21-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Binomial Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For 6 fair coin tosses, P(exactly 3 heads) is

A. 6/64

B. 3/6

C. 15/64

D. 20/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

788

QUESTION ID

S21-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20/64

GRADING NOTE C(6, 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

Why does C(n, k) count the outcomes with exactly k successes in n trials?

A. C(n, k) = n · k, the number of trials times the number of successes, counts every outcome with k successes.

B. Each such outcome is determined by choosing which k of the n trials are successes, and there are C(n, k) such choices.

C. The successes must come first, so there is exactly C(n, k) = 1 way.

D. Each of the trials has 2 outcomes, success or failure, so there are C(n, k) = 2k outcomes with k successes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

789

QUESTION ID

S21-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Each such outcome is determined by choosing which k of the n trials are successes, and there are C(n, k) such choices.

GRADING NOTE Choosing the positions of the successes. The others use false formulas for C(n, k).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

Rewrite rational expressions by polynomial division

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Write a division with a remainder as a mixed number	**4.NF.B.3**
BACKWARD 2	Multiply to check a division	**HSA-APR.A.1**
BACKWARD 3	Use the Remainder Theorem to check the remainder	**HSA-APR.B.2**
FORWARD 1	Divide with unknown coefficients and higher-degree divisors	**HSA-APR.D.6**
FORWARD 2	Use division to find slant asymptotes	**HSF-IF.C.7.d**

SET 22 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Which shows $(x^3 + 2x^2 - 5x + 3)/(x - 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^2 + x - 6 + 9/(x - 1)$

B. $x^2 + 3x - 2 + 1/(x + 1)$

C. $x^2 + 3x - 2 - 1/(x - 1)$

D. $x^2 + 3x - 2 + 1/(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

792

QUESTION ID
S22-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. $x^2 + 3x - 2 + 1/(x - 1)$

GRADING NOTE Synthetic division by 1: 1, 3, −2, remainder 1. The others divide by −1 instead of 1 ($x^2 + x - 6$, remainder 9), change the sign of the remainder, or change the divisor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

Which shows $(2x^3 - 3x^2 + 4x - 7)/(x^2 + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $2x - 3 + (2x + 4)/(x^2 + 1)$

B. $2x - 3 + (2x - 10)/(x^2 + 1)$

C. $2x - 3 + (2x - 4)/(x^2 + 1)$

D. $2x - 3 + (6x - 10)/(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

793

QUESTION ID

S22-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C.

$2x - 3 + (2x - 4)/(x^2 + 1)$

GRADING NOTE $(x^2 + 1)(2x - 3) = 2x^3 - 3x^2 + 2x - 3$; subtracting from the dividend leaves $2x - 4$ (degree less than 2). The others add instead of subtracting, or make a sign error in one term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

Which shows $(x^4 - 16)/(x - 2)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

B. $x^3 + 2x^2 + 4x + 8 - 32/(x - 2)$

C. $x^3 - 2x^2 + 4x - 8$ (remainder 0)

D. $x^3 + 8$ (remainder 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

794

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

GRADING NOTE Synthetic division by 2 with coefficients 1, 0, 0, 0, −16: 1, 2, 4, 8, remainder 0. The others divide by x + 2, leave out the zero placeholders, or miscompute the remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

Which shows $(6x^2 + x - 4)/(2x + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $6x - 2 - 3/(2x + 1)$

B. $3x - 1 - 4/(2x + 1)$

C. $3x - 1 + 3/(2x + 1)$

D. $3x - 1 - 3/(2x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

795

QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D.

3x − 1 − 3/(2x + 1)

GRADING NOTE $(2x + 1)(3x - 1) = 6x^2 + x - 1$, so the remainder is $-4 - (-1) = -3$. The others change the sign of the remainder, use synthetic division by $-1/2$ without dividing the quotient by 2, or forget to subtract the -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

Which shows $(x^3 - 4x + 5)/(x + 3)$ written in the form $q(x) + r(x)/b(x)$?

A. $x - 7 + 26/(x + 3)$

B. $x^2 - 3x + 5 - 10/(x + 3)$

C. $x^2 - 3x + 5 + 10/(x + 3)$

D. $x^2 + 3x + 5 + 20/(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

796

QUESTION ID

S22-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B.

$x^2 - 3x + 5 - 10/(x + 3)$

GRADING NOTE Synthetic division by -3 with coefficients 1, 0, -4, 5: 1, -3, 5, remainder -10. The others leave out the 0 for x^2, divide by 3 instead of -3, or change the sign of the remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 1

Write a division with a remainder as a mixed number • Grade 4 • Number & Operations–Fractions • 4.NF.B.3

Write 23/4 as a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

798

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 3/4

GRADING NOTE Any equivalent form (5 + 3/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 47/5 in the form q + r/5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

799

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9 + 2/5

GRADING NOTE Form required: q + r/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

17/3 = 5 + 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 17 = 3 · 5 + 2, so 17/3 = 5 + 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800